

Ausgrid Network Pricing Proposal For the Financial Year Ending June 2014

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Table of Contents

PART 1 APPROACH TO PRICE SETTING

1. INTRODUCTION AND OVERVIEW.....	4
2. BACKGROUND	6
3. PRICING METHODOLOGY.....	12
4. PROPOSED NETWORK PRICES FOR FY 2013/14	18
5. SMART GRID SMART CITY INITIATIVE	22
6. GLOSSARY	24

PART 2 DEMONSTRATING COMPLIANCE

1. OVERVIEW AND OUTLINE	27
2. NETWORK TARIFF CLASSES	28
3. PROPOSED TARIFFS AND CHARGING PARAMETERS.....	32
4. WEIGHTED AVERAGE REVENUE	45
5. VARIATIONS TO TARIFFS	50
6. CLIMATE CHANGE FUND	51
7. TRANSMISSION USE OF SYSTEM SERVICES	52
8. CHANGES FROM THE PREVIOUS REGULATORY YEAR	55
9. PUBLIC LIGHTING SERVICE	56
10. PROPOSED NEW NETWORK TARIFFS.....	58
11. REASONABLE ESTIMATES	59
12. ATTACHMENTS	61

Ausgrid FY14 Network Pricing Proposal

Part 1 Approach to Price Setting



1. Introduction and Overview

Introduction

Ausgrid's transmission and distribution services are subject to price regulation. The AER makes a five yearly determination, which sets a limit on the increase in distribution and transmission prices (X-factors) that can be applied by Ausgrid. The price limit is based on providing Ausgrid with a revenue requirement that is sufficient to recover efficient capital and operating costs. Ausgrid's X-factors for the 2009-14 regulatory period are set out below.

Table 1: Ausgrid's 2009-14 X Factors

	FY10	FY11	FY12	FY13	FY14
Forecast CPI	2.47%	2.47%	2.47%	2.47%	2.47%
Distribution	-17.86%	-18.18%	-18.18%	-18.18%	0.77
Transmission	-7.77%	-18.46%	-18.46%	-18.46%	-2.02

In addition to its decision on the X-factors to apply to distribution and transmission services, the AER must also make a decision on the price control to apply to public lighting services. The AER's determination for Ausgrid's public lighting services was published on 15 April 2010 and is discussed in Chapter 7 of Part 1 and the customer bill information and proposed prices for public lighting services are provided in Attachment B and C, respectively.

As a result of a new overall X factor increase transmission charges in FY14, under recovered transmission charges in FY13 and the contribution to the NSW State Government's Climate Change Fund, the total impact on the overall network component of the annual electricity bill for customers on the inclining block tariffs (IBT) in FY14 is summarised below:

- a typical residential customer¹ on IBT is estimated to receive a nominal bill increase of \$0.40 per week in FY14.

¹ For an average residential customer on the inclining block tariff who consumes 5,300 kWh per annum, and does not have controlled load hot water. Increase excludes GST. Retail related cost increases not included.

- a typical small business customer² on IBT is estimated to receive a nominal bill increase of \$0.85 per week in FY14.

These increases were offset by reductions in distribution prices as outlined later in this proposal.

Document structure

This document is Ausgrid's Pricing Proposal for FY14. It sets out how we propose to set network tariffs in FY14 to recover the efficient cost of owning, operating, maintaining and investing in the distribution and transmission networks.

This document also provides external stakeholders with the proposed prices for Ausgrid's alternative control (public lighting) services for FY14.

² For an average small business customer on the inclining block tariff who consumes 11,000 kWh per annum. Increase excludes GST. Retail related cost increases not included.

This proposal is submitted for review and approval by the AER as required by clause 6.8.2(a)(1) of the transitional chapter 6 of the National Electricity Rules (the Rules).

The proposal is split in two parts:

- Part 1 outlines Ausgrid's method and strategy in setting network prices. It is written with customers and other interested parties in mind, to assist in understanding how network tariffs are calculated.
- Part 2 is more technical in nature as it has been written specifically to demonstrate that Ausgrid's proposed network prices for FY14 comply with the obligations of the Rules. It also contains a discussion on Ausgrid's public lighting prices for FY14. Part 2 is structured to allow ready assessment of compliance by the AER.

While every attempt has been made to write Part 1 of this document in a style easily understood by external stakeholders that are not familiar with network pricing and regulatory concepts, it has been necessary to introduce the reader to some technical terms. To assist in understanding these technical terms, a comprehensive glossary section has been added to the end of Part 1.

2. Background

This section aims to provide an understanding of the historical context of Ausgrid's network prices for FY14 by discussing the progress made in reforming network tariffs over the past five years, the recent regulatory decisions by the Australian Energy Regulator (AER) for distribution and transmission network revenue and prices, and variations to this decision ordered by the Australian Competition Tribunal.

Ausgrid's distribution and transmission services are subject to price regulation by the AER under the Rules known as the 'Transitional Chapter 6 Rules'. The Rules apply for the 2009-14 regulatory period and are set out in Appendix 1 of Chapter 11 of the National Electricity Rules made under the National Electricity Law.

On a five yearly basis, the AER makes a determination which sets a limit on the price increases that we can charge customers (referred to as an X-factor for each year). The X-factors enable Ausgrid to recover an annual revenue requirement. The revenue requirement is based on the AER's appraisal of the efficient costs faced by Ausgrid for the five year period. As discussed below, distribution and transmission services are subject to different types of price control, and each have a set of X-factors.

Weighted Average Price Cap (WAPC)

Ausgrid's distribution prices are governed by a Weighted Average Price Cap (WAPC). The WAPC is a price cap where a limit is applied to the annual movement in distribution prices. The limit applied under either a revenue or price cap is referred to as the X factor and could be positive (price or revenue reduction) or negative (price or revenue increase), refer to Section 3 of Part 1 of this document for a more detailed explanation of the WAPC constraint.

Under the WAPC, distributors need to ensure that their distribution charges are as cost reflective as possible. Otherwise, distribution prices may not generate sufficient revenue to cover the efficient cost of distribution service provision in the event of unanticipated growth in peak demand for network capacity, which drives the need for additional investment in network augmentation.

By improving the efficiency of network tariffs, customers are likely to better understand the economic cost of their network use. To the

extent that customers do not value their network use more than the economic cost, customers will respond by changing their network use to eliminate this economic waste. According to economic theory, this type of customer response promotes the optimal allocation of society's scarce resources. One consequence of the WAPC is that Ausgrid is exposed to distribution revenue risk arising from unanticipated variations in electricity consumption, which is discussed below.

Revenue Cap

Ausgrid's transmission service prices are governed by a revenue cap. A revenue cap limits the amount of revenue that can be earned each regulatory year. This form of regulation provides Ausgrid with certainty that actual transmission costs incurred will be recovered from network customers, as any under (over) recovery of these transmission costs in a given year is allowed to be recovered in the following year (plus interest accrued) by adjusting transmission use of system charges in the subsequent period.

Electricity Consumption Trends

Every five years, the AER uses the distributor's forecast electricity volumes for the subsequent five year regulatory period to convert the allowable distribution revenue into a five year distribution price path. This in turn is used to calculate the cap on distribution price movements at each 1 July price reset (Table 1). The AER approves the forecasts and therefore plays a critical role in determining the level of risk faced by Ausgrid in earning its allowable distribution revenue over the five year regulatory period in question.

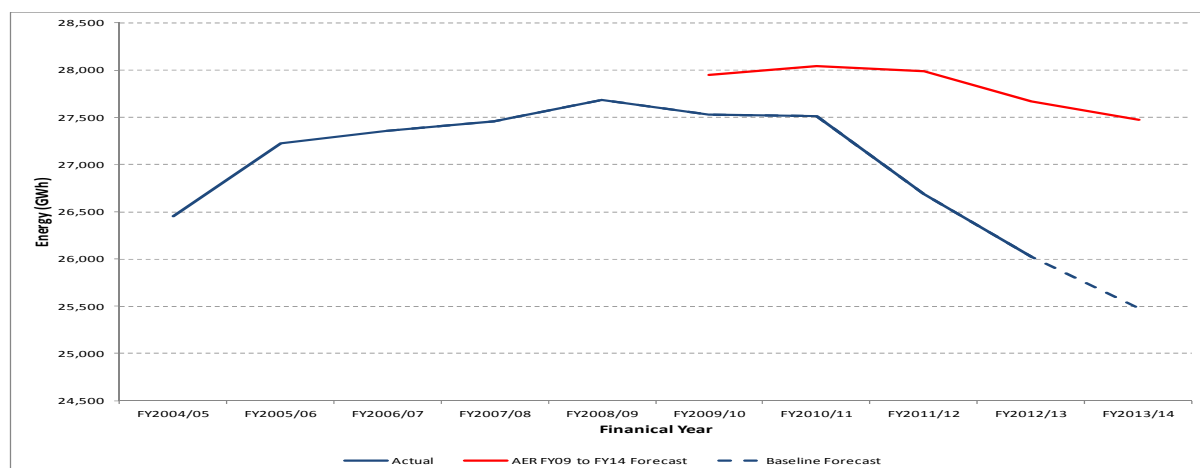
Importantly, Ausgrid is not only exposed to forecast volume risk at the time the X factors are set (every five years), but volume risk also arises from annual variation in electricity consumption volume due to such factors as unexpected weather, the take up of appliances such as air conditioners, or factors such as an economic downturn.

This volume risk translates into a distribution revenue risk because network prices are set on an annual basis, using the predetermined X factors, based on the most recent complete financial year of consumption. For example, the WAPC governing the setting of distribution tariffs in FY14 is based on the audited volumes

for FY12. In simple terms, this means that if actual consumption in FY14 is lower than the audited consumption in FY12, distribution revenue in FY14 will fall short of expectations.

A summary of the differences between the overall forecast volumes used to set the X

factor in the AER determination and a feasible range of forecasts based on more recent data is shown in the figure below.



The future uncertainty in Ausgrid's volume environment reflected in Figure 1 above highlights the need for Ausgrid to continue to improve the cost reflectivity of network tariffs. It also highlights the importance of continued research into the underlying drivers of the electricity consumption behavior of households and business in Ausgrid's network area.

Ausgrid is keen, noting the historical price outcomes over the preceding four years, to do all it can to keep prices close to CPI in the future. To achieve this in FY2013/14 Ausgrid proposes to reduce DUOS prices on average by 1.5% under the WAPC, rather than increase DUOS prices on average by 1% as allowed under the current determination. This proposed DUOS price reduction under the WAPC is necessary to ensure that our published tariff customers receive an increase at the NUOS level in FY14 close to CPI.

Over the next twelve months, Ausgrid intends to undertake a further review of tariff and metering strategy with the purpose of developing a more equitable and efficient approach to pricing in the 2014-19 determination period. This review is required due to the AER decision to move from a WAPC to a revenue cap from 1 July 2014. Ausgrid will inform stakeholders of the results of this review as part of the consultation process for the FY 2015 prices.

Ausgrid's 2009-14 X -Factors

On 25 November 2009 the Australian Competition Tribunal made orders varying the AER's distribution determination for 2009–14.

The Tribunal's determination varied the X factors applied to Ausgrid's distribution and transmission services for the last four years of the regulatory period. The X factors for Ausgrid's distribution and transmission services during the 2009–14 regulatory period are set out below, in addition to the respective annual revenue requirements

Table 2: Australian Competition Tribunal Outcome for Ausgrid

Ausgrid	FY10	FY11	FY12	FY13	FY14
Distribution X factors	-17.86%	-18.18%	-18.18%	-18.18%	0.77%
Transmission X factors	-7.77%	-18.46%	-18.46%	-18.46%	-2.02%
Annual revenue requirement (distribution) nominal \$m	1,330.5	1,519.6	1,702.8	1,903.1	2,059.3
Annual revenue requirement (transmission) nominal \$m	146.7	182.8	206.6	238.4	273.6

Public Lighting Services

Ausgrid's public lighting services regulated by the AER through a price control mechanism that is separate to that for distribution standard control services. It is reviewed every five years alongside the review of the distribution price control. The most recent regulatory review concluded in April 2009, when the AER released its final determination for distribution and transmission.

On 25 November 2009 the Australian Competition Tribunal made orders varying the AER's distribution determination for public lighting charges. The AER's determination is on its website and chapter 9 provides an overview of the prices and charges that will result from that determination.

TransGrid

Ausgrid's network prices not only recover the efficient cost of operating, maintaining and augmenting Ausgrid's distribution and transmission assets, but they also include recovery for a portion of TransGrid's transmission costs.

TransGrid is the owner, operator and manager of the NSW high voltage network, connecting generators, distributors and major end users in New South Wales. TransGrid's network comprises almost 12,500 kilometres of high voltage transmission line and underground cables up to a voltage of 500kV. It also has six interconnectors to Victoria and Queensland's transmission network. 60 percent of NSW energy is generated west of the Great Dividing Range and it must be delivered to the east coast, where most of the state's load is located. The Sydney, Wollongong and Newcastle areas consume 75 percent of the state's energy.

TransGrid also received a five-year revenue determination from the AER in April 2009. TransGrid operates under a revenue cap, which provides more revenue certainty than the price cap governing Ausgrid's distribution prices. TransGrid also sought review of elements of the AER's determination, and as a consequence the Australian Competition Tribunal decided to vary the maximum allowed revenue (MAR) that was set out in the AER's April 2009 determination for TransGrid. The smoothed MAR for TransGrid is set out below.

Table 3: Australian Competition Tribunal Outcome for TransGrid

TransGrid	FY10	FY11	FY12	FY13	FY14
TransGrid Maximum Allowed Revenue nominal \$m	678.4	734.2	794.5	859.8	930.5

Some of the reasons for the recent significant increases in TransGrid's revenue recovery are as follows:

- increases in the regulated revenue caps for NSW transmission businesses for the next pricing period;
- a continuing fall in the value of National Electricity Market trading residues returned to customers as a reduction in transmission charges; and
- any pass through adjustments that have been approved by the AER.

In July 2010 TransGrid significantly rebalanced its transmission charges by removing the peak and shoulder energy prices and more than doubling the demand charge. Given the significance of this rebalancing of TransGrid's transmission charges, Ausgrid has decided to transition its transmission tariffs to cost reflective levels over a reasonable period where required to avoid imposing unacceptable price shocks on individual customers in the remaining years of the regulatory period.

More recently, TransGrid has announced a revenue freeze for FY 2013/14 as well as a commitment to limit increases to transmission charges to CPI in the 2014-19 determination period.

this scheme is explained in Part 2 of this document.

General Overview

The revenue stream for the Ausgrid network business is derived from our customer base in the proportions shown in Figures 2a.

Solar Bonus Scheme

The NSW Government introduced the Solar Bonus scheme (SBS) in January 2010 to encourage small-scale generation in NSW, such as a solar photovoltaic (PV) system or wind turbine. Under the SBS, eligible participants³ in this scheme are paid for the electricity generated. This scheme is closed to new connections. The network pricing arrangements associated with participation in

³ Customers using less than 160 MWh pa with a small-scale generation unit of a maximum capacity of 10 kilowatts.

Figure 2a: FY14 Expected Revenue by Customer Segment

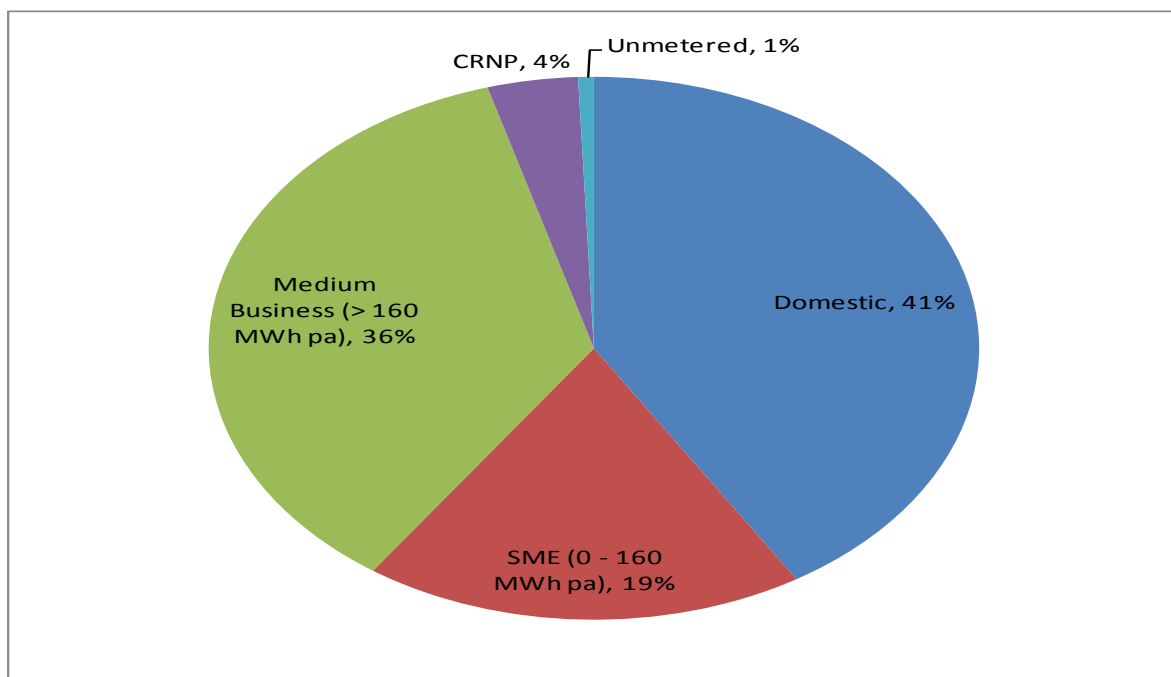
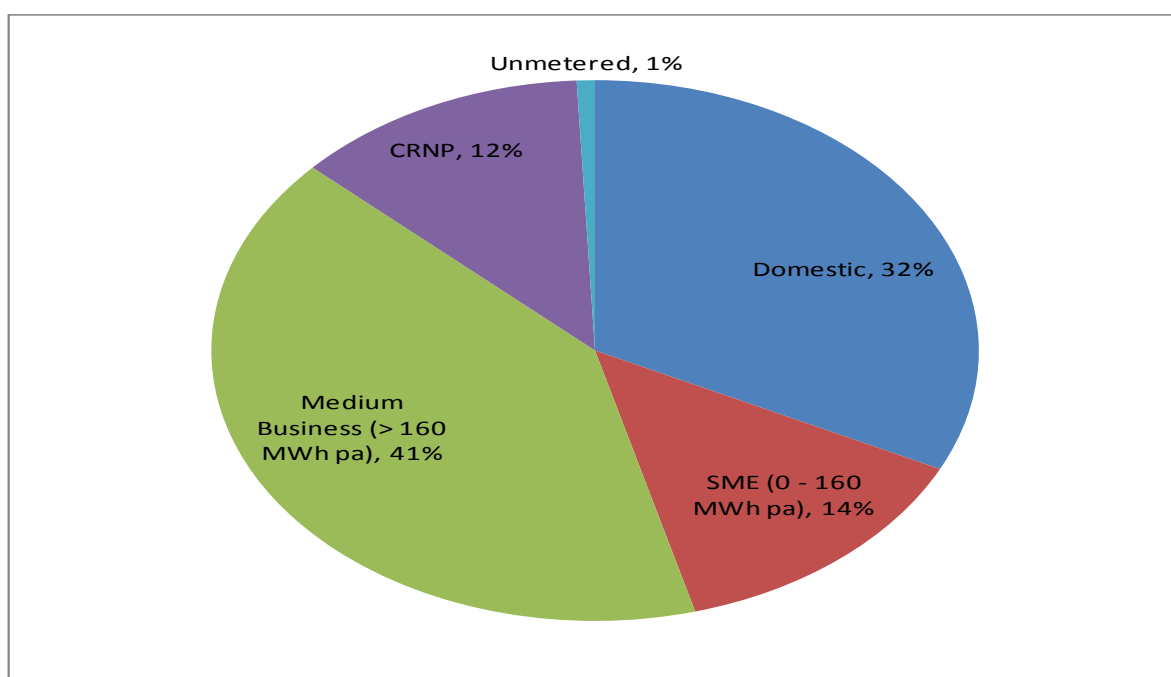


Figure 2b: FY14 Expected Energy Consumption by Customer Segment



As can be seen from the above figures, Ausgrid's largest customers on a Cost Reflective Network Price (CRNP) tariff represent 12% of energy volumes but contribute only 4% of revenues. Conversely, residential customers consume 32% of the

total energy volumes, but account for 41% of revenue. This is reflective of the amount of assets required to service these customers. For instance, large customers, though they use large volumes, are connected high up in the distribution network, whereas smaller

customers use the electricity network all the way down to their LV connection. Therefore the cost reflective price is higher for low voltage connected customers, compared to the cost reflective price for customers connected at either the high voltage or sub-transmission voltage level.

Finally, Ausgrid's network revenue is comprised of a number of components, as summarised in the following figure:

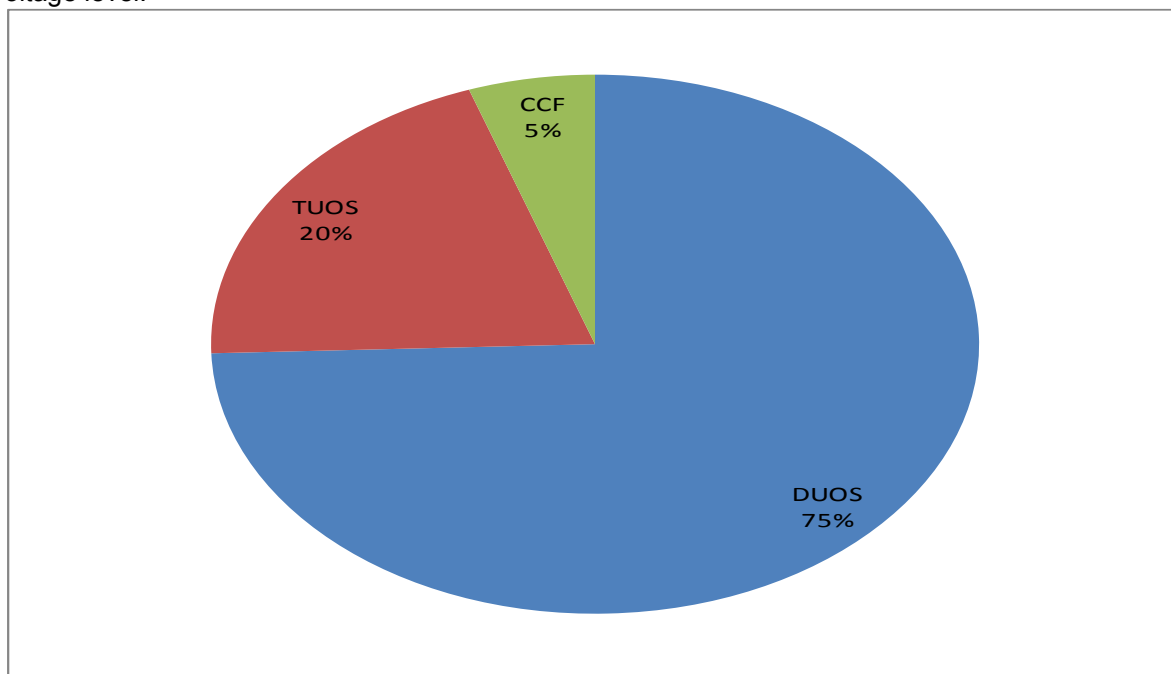


Figure 3: FY14 Expected Revenue by Revenue Source

As shown above, the cost of owning, operating, maintaining and investing in the distribution network is the largest component of annual network revenue accounting for around 75%. Included in the distribution revenue are earnings derived from a range of ancillary activities, such as extra meter readings, new connections, answering conveyancing enquires etc. This accounts for approximately 1% of distribution revenue.

The second largest component is the recovery of the cost of transmission network services⁴, which accounts for around 20% of annual network revenue. Ausgrid is also required to make a contribution to the NSW Government Climate Change Fund, which is used to promote energy efficiency and renewable energy technology. This contribution accounts for only 5% of annual network revenue.

⁴ The transmission network connects large generators together, and transports the energy they produce to distributors.

3. Pricing Methodology

The following discussion outlines how Ausgrid sets network charges, by providing an understanding of the four basic steps in setting of network tariffs, as summarised below:

- Identify a range of objectives for network tariffs to achieve.
- Develop pricing strategies that aim to achieve these objectives in a manner that strikes an appropriate balance between financial, economic and equity considerations.
- Establish an equitable amount of revenue to recover from each tariff class based on the user pays principle.

- Set network tariffs in accordance with the above steps and in compliance with price limits and the weighted average price cap imposed by AER.

Pricing Objectives

The network prices are set each year to achieve a range of economic and non-economic objectives, as summarised in the following table.

Table 4: Ausgrid's Network Pricing Objectives

Objective		Description/Principle
Achieve economic efficiency by providing customers with prices reflective of economic cost		Efficient network tariffs signal to retailers the economic cost of providing network capacity to end-customers at different times.
Achieve regulatory outcomes		Network tariffs will be set in accordance with the National Electricity Rules and the AER determination. This means that distribution prices should, depending on volume considerations, generate revenues that cover the efficient cost of owning, maintaining, operating and augmenting the network.
Satisfy customers	Comprehensible	Network tariffs should be designed to: • Be as simple as possible so that customers are able to understand the bill implications of usage decisions. • Result in customers connected at the same level of the network with similar usage patterns receiving similar network charges. • Provide certainty of price movements, which in practice means that in some situations distribution prices need to be transitioned to desired levels to avoid imposing unacceptable price shocks on customers.
	Equitable	
	Certainty	
Equitable recovery of common network costs		Allocate common costs of network service provision to tariff classes on the basis of user pays principle to ensure that prices lie between avoidable and standalone cost.

The pricing objectives identified in the above table are discussed in more detail below:

Achieve economic efficiency

Ausgrid believes that it is important that network tariffs provide customers with price signals reflective of the economic cost of network service provision. This approach to setting network tariffs ensures that customers make economically rational decisions about their network use.

A key strategy for achieving economic efficiency is to set prices for the use of the electricity distribution network during periods when the network is more likely to be congested reflective of marginal cost. This approach ensures that customers are informed of the economic cost of their network use and are more likely to make economically rational decisions in relation to changing their behaviour or investing in more energy efficient household appliances.

While some customers may place a high value on their peak network use and, therefore, be willing to pay the higher price of network use during peak periods, other customers are more likely to respond to these peak price signals by minimising their use of network during periods of network congestion, which is likely to contribute to a reduction in peak demand and consequently less need to invest in network peak capacity.

A full discussion on economic network pricing concepts is found in Part 2 of this document.

Achieve regulatory outcomes

To ensure the long-term viability of the network, regulators establish the annual revenue requirement of the distribution business by considering the efficient costs of owning, operating, maintaining and investing in the network over the regulatory period. This distribution revenue requirement is translated into X factors that cap the movement in weighted average price in each year of the regulatory period. This type of price control is referred to as a price cap.⁵

Under a price cap, the distributor is exposed to the risk that its tariffs will not generate sufficient distribution revenue to cover the efficient cost of owning, operating, maintaining and augmenting its distribution network. Therefore, an important aspect of the network price setting process is to develop strategies to

deliver regulatory outcome in a manner that does not expose the business to unacceptable levels of risk that distribution revenue will not be recovered. In an economic and volume environment of considerable uncertainty, this represents a significant challenge as it is important that revenue risk mitigation is achieved without compromising other pricing objectives, such as equity and efficiency.

Make prices comprehensible

Ideally, all customers should be able to understand how prices are determined and how their bills are calculated. Unfortunately, simplicity is not always consistent with achieving other goals. Developing prices which are as equitable as possible, which encourage rational economic behaviour and which ensure the right return on investment would lead to quite complex price structures. How the actual prices are calculated for each charging parameter (eg network access charge) is also similarly complicated given the competing objectives underpinning the price setting process. A major aim of this document is to provide a broad understanding of the process used to develop network prices.

More complex network prices have tended to apply to larger customers such as businesses, while residential and small business customers – who comprise over 90% of the total customer base - have had fairly simple price structures. However, to continue to provide small customers with a highly averaged pricing signal is not desirable on economic and equity grounds. Over the remainder of the regulatory period, Ausgrid will continue to focus on exploring a range of reform options that aim to lay the foundation for more cost reflective pricing signals for all network customers. The challenge will be to develop pricing structures that are not only more cost reflective, but are also able to be understood by retailers and their end-customers.

Provide certainty over future price trends

It is not desirable to make unexpected changes to the structure and level of network charges as this is likely to undermine the willingness of customers to make permanent changes in their network use, such as investing in more energy efficient appliances. Changes to the structure of tariffs creates uncertainty over the extent to which these changes result in lower network bills in the future.

It is for this reason that Ausgrid is committed to providing external stakeholders with as much certainty as possible over the future direction

⁵ Unlike distribution, the recovery of transmission costs is governed by an overs and unders account ie revenue cap.

of network prices. This commitment takes many forms, such as publishing the pricing proposal and statement of expected price trends, stakeholder engagement and consultation that aim to provide interested parties with an understanding of Ausgrid's approach to setting network tariffs and potential tariff reform options being investigated.

This commitment also directly influences the setting of network tariffs to ensure that prices are transitioned to cost reflective levels, where necessary to avoid imposing unexpected price shocks on customers. Ausgrid is confident that this approach will realise the benefits of more efficient prices without raising undue equity concerns.

Make prices equitable for all customers

Ausgrid seeks to ensure that network prices are equitable as well as economically efficient. This concept of equity means that two customers with similar network usage patterns located next door to each other should expect to be charged on the same basis, customers whose location or consumption pattern cost the DNSP more to service should expect to pay a greater cost.

Equitable recovery of common network costs

Ausgrid believes that the allocation of common network costs to individual tariff classes on the basis of the user pays principle is the most equitable approach to establishing how much distribution revenue to recover from each tariff class. In recent years, Ausgrid has been refining its cost allocation approach so that the amount of distribution revenue recovered from a particular tariff class is as close as possible to the cost incurred by Ausgrid to provide network service to these customers. This approach will also ensure that distribution prices are likely to lie between avoidable and standalone cost, as required by the National Electricity Rules (as discussed in detail in Part 2 of this document). This is important as it means that the network tariff setting process will not result in economic subsidies between tariff classes.

While it is desirable in theory to set network tariffs that achieve perfect cost reflectivity at the tariff class level, in practice such an approach would result in unnecessary pricing uncertainty given that the cost allocation process is influenced by relative volume movements. In practice, it is preferable to balance the competing needs of price path

certainty and the user pays principle by transitioning tariffs to cost reflective levels over a reasonable time period. For example, Ausgrid has adopted transitional network pricing arrangements to avoid imposing unacceptable bill impacts on customers from directly passing through TransGrid's transmission charges in FY14.⁶

⁶ Note that TransGrid significantly reformed the structure of transmission charges in FY11 by removing the peak and shoulder energy charges and increasing demand charges by around 150%.

Table 5: Summary of Ausgrid's Network Tariff Strategy

Objective		Strategy/Comment
Achieve economic efficiency by providing customers with prices reflective of economic cost		Network tariffs have been designed to reflect economic cost to the extent possible given existing tariff structures and metering constraints. Ausgrid is also committed to investigating more cost reflective forms of network pricing, such as dynamic tariffs and more innovative approaches to signaling economic cost, such as dynamic rebates and to increasing the number of customers on more cost reflective network tariffs, where it is economically and socially desirable to do so. It is for this reason that Ausgrid proposes to re-assign a significant number of customers to a more cost reflective network tariff in FY14 given their consumption and metering functionality.
Achieve regulatory entitlements		Network tariffs have been developed in accordance with the Weighted Average Price Cap constraint and price limits and the pricing principles set out in the National Electricity Rules (refer to Part 2 of this document). As a result of increased uncertainty about future volume movements, greater emphasis has been placed on increasing those tariff components that are less sensitive to volumes where this is economically justified and socially desirable to do so.
Satisfy customers	Comprehensible	Ausgrid is committed to simplifying the network tariffs where appropriate.
	Fairness	Ausgrid is committed to developing network tariffs and assigning customers to network tariffs on a non-discriminatory basis (refer to Attachment G).
	Stable	Ausgrid is committed to provide retailers and end-customers with certainty over the direction of network prices through publishing pricing documents, the retailer's forum and through the establishment of transitional price path arrangements to cost reflective levels.
Equitable recovery of common network costs		Ausgrid is committed to improving its allocation of network costs to individual tariff classes to ensure that the amount of distribution revenue recovered from customers in a particular tariff class is as close as possible to the costs that Ausgrid incurs on these customers' behalf.

Ausgrid believes that it is important to provide external stakeholders with an understanding of not only the basis of the proposed network prices for FY14, but also the potential medium to longer term direction of Ausgrid's network prices. In this way, retailers and other interested parties will be better aware of the potential reforms to network tariff structures and weightings between charging parameters. The following discussion provides an understanding of the key pricing initiatives that are most likely to influence the future direction of Ausgrid's network prices:

Ausgrid's Time of Use (ToU) electricity pricing program

Ausgrid has been involved in delivering innovative Time of Use (ToU) metering and pricing solutions in Australia since 2004 with its ToU program, as summarised below:

- Disc meters at the majority of medium sized customer premises (in the consumption range of 15-160 MWh per annum) have been replaced by interval meters. This meter replacement program has reached completion;
- Most customers with interval meters have been transferred to ToU prices; and
- All new and upgraded customer connections (residential dwellings and businesses) receive an interval meter with an option to move to a ToU tariff.

In its 2004 determination, the Independent Pricing and Regulatory Tribunal (IPART), facilitated this ToU program by allowing expenditure for meter replacement and modifying the pricing side constraints to allow ToU prices to be mandated.

As at 30 June 2012, 325,813 homes and 90,010 businesses had time of use meters and were on time based pricing in the Ausgrid network area.

Ausgrid has recently made the time of use tariff a voluntary option for residential customers, reflecting our commitment to providing customers greater choice over their network tariff. Ausgrid will, however, continue to install interval meters in all new connections, but these customers will be assigned to the Inclining Block Tariff.

Smart Grid Smart City Initiative

Ausgrid is currently testing innovative tariffs as part of the *Smart Grid Smart City initiative*. More information on this project and the other important research projects undertaken by Ausgrid is found on our website www.ausgrid.com.au.

Compliance with regulatory constraints

DUOS

The AER's 2009-14 determination specifies a Weighted Average Price Cap (WAPC) control mechanism for standard (distribution) control services. The WAPC limits price movements over the regulatory period, so that the business receives revenue gains or losses depending if volumes rise or fall. The WAPC works by

weighting each proposed price by its historic volumes, and dividing this result by existing prices weighted by the same historic volumes. This "weighted average" price increase is limited to a combination of CPI, X factor and D factor for that year. The weighted average price cap formula is defined below:

Figure 4: Weighted Average Price Cap (WAPC) Formula

$$\frac{\sum_{i=1}^m \sum_{j=1}^n p_{t-1}^{ij} \times q_{t-2}^{ij}}{\sum_{i=1}^m \sum_{j=1}^n p_{t-1}^{ij} \times q_{t-2}^{ij}} \leq (1 + \Delta CPI)(1 - X_t)(1 + D_t)$$

where:

- p_t = the proposed prices for the upcoming financial year
- p_{t-1} = prices for the current financial year
- q_{t-2} = volume quantities from the most recently completed financial year
- X_t = the allowed real average price increase set by the AER from their April 2009 Determination
- D_t = the AER approved adjustment to recover costs from successful demand management initiatives
- ΔCPI = the most recent historic average December to December change in CPI

For FY14, the change in CPI is 1.76%, the allowed X-factor is 0.77% and the D-factor is 0.10%. Distribution prices are therefore allowed to increase on a weighted average basis by 1.08% for FY14.

Ausgrid proposes to reduce DUOS prices on average by 0.9% under the WAPC in FY14, rather than increase these prices by the allowed increase of 1.08% under the determination. This proposed reduction in DUOS prices under the WAPC ensures that our published tariff customers receive an increase at the NUOS level in FY14 close to CPI. Refer to Part 2 of this document for a comparison of the overall WAPC constraint and the weighted average distribution price increase for FY14.

Price Limits

In addition to the constraint under the Weighted Average Price Cap, increases to Ausgrid's distribution prices are also constrained by price limits that apply to each tariff class, rather than on an individual tariff basis. Specifically, the National Electricity Rules require that the expected weighted average distribution revenue to be raised from a tariff class must not exceed the overall CPI+X constraint plus 2%

TUOS

The AER's 2009-14 Determination for standard control distribution services requires that Ausgrid set TUOS prices each year to achieve a forecast zero closing balance of the Transmission Overs and Unders account in the year that the proposed TUOS prices apply. In other words, Ausgrid's proposed TUOS prices must be set to recover the following:

- The allowed revenue relating to Ausgrid's transmission services, as per the AER's 2009-14 Transmission Determination.
- The forecast cost of transmission services to be provided by TransGrid, based on the transmission charges and forecast usage applying to TransGrid-owned transmission connection points in Ausgrid's network area.
- Any forecast under/over recovery of Ausgrid's TUOS revenue in the previous year (including interest).

For a more detailed explanation of our TUOS pricing arrangements, please refer to Part 2 of this document.

CCF

Ausgrid is required to set the CCF component of network tariffs each year to achieve a forecast zero closing balance of the Overs and Unders account in the year that the proposed CCF prices apply. In other words this means that the proposed CCF prices must be set to

recover Ausgrid's annual contribution to the CCF and any forecast under/over recovery from the previous year (including any interest)

For the year ending June 2014, Ausgrid is required to contribute \$149.8M to the Climate Change Fund. Ausgrid expects to recover \$149.7M for this purpose from network charges

For a more detailed explanation of our CCF pricing arrangements, please refer to Part 2 of this document.

4. Proposed Network Prices for FY 2013/14

This section aims to provide external stakeholders with an understanding of the proposed changes to the major published network tariffs (inclusive of Climate Change Fund contribution) for FY14. Please note that the prices presented below have been rounded to two decimal places for presentation purposes.⁷

The following table provides a comparison of the network tariffs (inclusive of Climate Change Fund contribution) for FY14 and the proposed network tariffs for FY14 for residential customers.

Table 6: Proposed Network Tariffs – FY14 – Residential Customers

Charging Parameter	Actual FY13	Proposed FY14	% Variation
Inclining Block Tariff (EA010)			
Fixed Charges (c/day)	38.00	38.95	2.5%
First Block (c/kWh)	12.60	12.92	2.5%
Second Block (c/kWh)	15.00	15.38	2.5%
Third Block (c/kWh)	19.00	19.48	2.5%
Time of Use Tariff (EA025)			
Fixed Charges (c/day)	50.00	51.25	2.5%
Peak (c/kWh)	25.50	26.14	2.5%
Shoulder (c/kWh)	5.02	5.15	2.5%
Off-peak (c/kWh)	2.56	2.62	2.5%
Note - Prices exclude GST.			

⁷ Please refer to chapter 3 of Part 2 of this document for a complete listing of Ausgrid's proposed prices for FY14 for all of the published network tariffs.

The following table provides a comparison of the actual network tariffs for FY13 and the proposed network tariffs for FY14 for controlled load customers.

Table 7: Proposed Network Tariffs – FY14 – Controlled Load Customers

Charging Parameter	Actual FY13	Proposed FY14	% Variation
Controlled Load 1 Tariff (EA030)			
Fixed Charges (c/day)	3.16	3.24	2.5%
Non-TOU Charge (c/kWh)	2.10	2.15	2.5%
Controlled Load 2 Tariff (EA040)			
Fixed Charges (c/day)	12.22	12.53	2.5%
Non-TOU Charge (c/kWh)	5.20	5.33	2.5%
Note – Prices exclude GST.			

The following table provides a comparison of the actual network tariffs for FY13 and the proposed network tariffs for FY14 for small business customers.

Table 8: Proposed Network Tariffs – FY12 – Small Business Customers

Charging Parameter	Actual FY13	Proposed FY14	% Variation
Inclining Block Tariff (EA050)			
Fixed Charges (c/day)	120.00	123.00	2.5%
First Block (c/kWh)	11.24	11.52	2.5%
Second Block (c/kWh)	19.74	20.23	2.5%
Time of Use Tariff (EA225)			
Fixed Charges (c/day)	120.00	123.00	2.5%
Peak (c/kWh)	20.60	21.12	2.5%
Shoulder (c/kWh)	6.39	6.55	2.5%
Off-peak (c/kWh)	2.00	2.05	2.5%
Note – Prices exclude GST.			

The following table provides a comparison of the actual network tariffs for FY13 and the proposed network tariffs for FY14 for medium/large business customers connected at the low voltage level of the distribution network.

Table 9: Proposed Network Tariffs – FY14 – Medium Businesses

Charging Parameter	Actual FY13	Proposed FY14	% Variation
Time Of Use Tariff 40MWh to 160 MWh pa (EA302)			
Fixed Charges (c/day)	550.00	563.75	2.5%
Peak (c/kWh)	13.78	14.12	2.5%
Shoulder (c/kWh)	6.28	6.44	2.5%
Off-peak (c/kWh)	3.78	3.87	2.5%
Capacity (c/kW/day)	33.00	33.83	2.5%
Time Of Use Tariff 160MWh to 750 MWh pa (EA305)			
Fixed Charges (c/day)	1600.00	1640.00	2.5%
Peak (c/kWh)	10.84	11.11	2.5%
Shoulder (c/kWh)	5.84	5.99	2.5%
Off-peak (c/kWh)	2.84	2.91	2.5%
Capacity (c/kVA/day)	33.00	33.83	2.5%
Time Of Use Tariff Above 750 MWh pa (EA310)			
Fixed Charges (c/day)	1999.00	2048.98	2.5%
Peak (c/kWh)	11.01	11.29	2.5%
Shoulder (c/kWh)	5.81	5.96	2.5%
Off-peak (c/kWh)	2.81	2.88	2.5%
Capacity (c/kVA/day)	33.00	33.83	2.5%
Note – Prices exclude GST.			

The following table provides a comparison of the actual network tariffs for FY11 and the proposed network tariffs for FY12 for large business customers connected to the high voltage level of distribution network.

Table 10: Proposed Network Tariffs – FY14– High-Voltage Connected Customers

Charging Parameter	Actual FY13	Proposed FY14	% Variation
High Voltage Time Of Use Capacity Tariff (EA370)			
Fixed Charges (c/day)	4000.00	4100.00	2.5%
Peak (c/kWh)	8.44	8.65	2.5%
Shoulder (c/kWh)	5.24	5.37	2.5%
Off-peak (c/kWh)	2.33	2.39	2.5%
Capacity (c/kVA/day)	17.30	17.73	2.5%
Note – Prices exclude GST.			

The following table provides a comparison of the actual network tariffs for FY13 and the proposed network tariffs for FY14 for large business customers connected to the sub-transmission voltage level of distribution network.

Table 11: Proposed Network Tariffs – FY14 – Sub-transmission Customers

Charging Parameter	Actual FY13	Proposed FY14	% Variation
Sub-transmission Voltage Time Of Use Capacity Tariff (EA390)			
Fixed Charges (c/day)	5000.00	5125.00	2.5%
Peak (c/kWh)	4.71	4.83	2.5%
Shoulder (c/kWh)	3.31	3.39	2.5%
Off-peak (c/kWh)	1.96	2.01	2.5%
Capacity (c/kVA/day)	5.54	5.68	2.5%
Note – Prices exclude GST.			

5. Smart Grid Smart City Initiative

Overview

Smart Grid, Smart City is an Australian government funded project, led by Ausgrid and supported by our consortium partners. The project is testing a range of smart grid technologies, gathering information about the economic and environmental benefits and costs of implementing these technologies in an Australian setting. It is expected that up to 30,000 households will participate in the project, which runs between 2010 and 2013.

Ausgrid is leading a consortium of partners working together on this project. EnergyAustralia, the *Smart Grid, Smart City* retailer partner, is testing a range of innovative technology and pricing offers, including new generation smart meters. These products are also designed to give energy consumers greater choice and control over their bills.

Building a smart grid also involves transforming the traditional electricity network by adding a chain of new smart technology. Technologies include smart sensors, new back-end IT systems, smart meters and a communications network. A range of technologies and products are being tested on both the electricity network and within households.

The focus of the trial is Newcastle, with initiatives also taking place in Newington, Sydney's CBD, Kuring-gai and Scone.

The consortium testing the smart grid technologies and ensuring their suitability for Australian conditions includes EnergyAustralia, Landis+Gyr, GE Energy, Grid Net, IBM Australia, Better Place, CSIRO, TransGrid, Hunter Water, Sydney Water, University of Newcastle, University of Sydney, Lake Macquarie City Council and Newcastle City Council.

Customer Applications

The *Smart Grid, Smart City* project involves a large trial of 'Customer Applications' - technology and tariff related products for residential consumers. The customer applications program is running both network trials, without tariff change, and opt-in retail trials with tariff change. This is in combination with different feedback technologies that test the effects of different information, pricing and technology configurations on customers' electricity usage over the long term and in peak events. Both streams test the feedback technology developed for *Smart Grid, Smart City*. The network trial has acquired participants and the retail trial is nearing the end of its customer acquisition phase. Elements of the Customer Applications project include:

1. Information feedback devices.
2. Community education programs eg. Residential energy audits.
3. Rebates from Networks and/or Retailers to reward demand management activities by consumers i.e. energy saving/shifting efforts.
4. Engineering solutions such as Air Conditioner cycling to manage network load.
5. Innovative tariffs that involve a Network and Retail component.

In keeping with common practice in the NEM, each retail tariff structure in item 5 is based on the underlying network tariff structure. Each tariff aims to deliver benefits in at least one of the following categories, with no negative impacts in the others:

- Consumers – through enhanced visibility of their energy use and likely costs, and a wider range of bill savings and payment options.
- The Environment – through reduction in energy usage.
- Networks – through directing consumer's energy usage away from network peaks.

- Retailers – through directing consumer energy usage away from periods of high wholesale costs.

The development of new tariffs is also limited by available technology. For example, while Real Time Pricing (RTP) has theoretical merit, the outcome of discussions with project stakeholders prioritised tariffs which are closer to having the requisite technologies in place.

The *Smart Grid, Smart City* project is mindful of the need to ensure suitable customer protection strategies are in place during the trial to ensure that no customer is disadvantaged. Elements of the trial designed to ensure this are:

- All *Smart Grid, Smart City* tariffs are voluntary and Retail contracts will not include penalties such as exit fees during the term of the trial.
- Providing a high standard of information to customers to allow them to choose options from which they can gain the most benefit.
- The *Smart Grid, Smart City* project has committed to working with community groups representing the interests of customers who may experience hardship due to current or future energy costs.

Operation of the EnergyAustralia Pricing and Technology Trial

The EnergyAustralia *Smart Grid, Smart City* pricing trial is testing a range of tariffs to give customers more control of their energy use and costs:

- 1 EnergyAustralia markets innovative Retail tariffs to its customers in the trial areas to opt-in to the trial. As the innovative tariffs and any feedback technology require time-based metering, all customers taking up an offer will have a smart meter installed.
- 2 The Retailer partner informs Ausgrid as to which customers took up which *Smart Grid, Smart City* Retail tariff product. Ausgrid will from that point charge the corresponding Network Tariff and the Retailer will commence billing the new Retail tariff.
- 3 Ausgrid will return each site to a non-*Smart Grid, Smart City* Network Tariff when the customer exits their *Smart Grid, Smart City* Retail tariff, for example, at the end of the trial period.

Tariff Design Principles

The design of the *Smart Grid, Smart City* network tariffs has been informed by local and international thinking on efficient and cost-reflective tariff structures that have potential to deliver environmental and economic benefits to customers, the environment, Networks and Retailers.

6.Glossary

Term	Definition
AER	Australian Energy Regulator
Bulk Supply Point (BSP)	A vital step in the transmission of electricity is decreasing the transmission line voltage using transformers. This is performed at a large substation called a Bulk Supply Point owned and operated in NSW by TransGrid
Capacity Charge	A capacity charge applies to the maximum half hourly kW or KVA recorded during the peak period of a working weekday over the 12 months prior to a bill being calculated
Controlled load Tariff	Applicable to electricity which is separately metered and controlled by Ausgrid and used for operating storage water heaters, thermal storage space heaters, and other approved fixed wired appliances
Cost Reflective Network Prices (CRNP) Tariff	A price calculated for customers using more than 10MW, that takes into account load flows of the specific assets the customer uses
Determination	2009 AER Determination for NSW Electricity Distribution Service Providers
Distribution Use of System (DUOS)	The revenue earned for the provision of distribution services
Dynamic Peak Rebate	This is a rebate offered to customers for consumption reductions during a dynamic peak event
Dynamic Tariff	This tariff applies a price for energy consumed during a dynamically signalled congestion period
Economic Efficiency	The improvement in societal welfare from providing customers with a price signal reflective of marginal, rather than average cost of distribution service provision.
Equity	The impact of price changes on the well being of disadvantaged groups
First Block	See <i>Inclining Block Tariff (IBT)</i>
Fixed Charge	This tariff component does not vary with electricity usage at the customer's premises and is charged on a cents/day/NMI basis
Inclining Block Tariff (IBT)	This tariff typically has a fixed charge and a variable energy charge component with a low price charged up to a predetermined consumption threshold (first block) and a different price charged to customers who consume electricity above predetermined thresholds (the second or third blocks)
Network Use of System (NUOS)	The revenue earned for the provision of transmission and distribution services
IPART	Independent Pricing and Regulatory Tribunal
Off-peak Consumption Charge	This tariff component applies to the price charged for electricity usage during the off-peak period of the day. See also <i>Time of Day Period Definition</i>
Overs & Unders Account (Transmission)	This account provides Ausgrid with certainty of transmission cost recovery by allowing future TUOS prices to increase (decrease) to recoup (return) an under (over)-recovery of TUOS
Peak Consumption Charge	This tariff component applies to the price charged for electricity usage during the peak period of the day. See also <i>Time of Day Period Definition</i>

Term	Definition
Price Cap	Refer to Weighted Average Price Cap
Real-time Tariff	This type of tariff has variable prices that vary every half-hour interval in response to change in energy purchase costs
Revenue Cap	A form of monopoly regulation where the regulated entity's allowed revenue from tariffs is limited by an X factor. Any over or under-recovered of actual revenue is accounted in an adjustment to tariffs in the subsequent period.
Revenue Requirement	The level of revenue that is adequate to fund the on-going operation of the business and provide a commercial return to the owner
Reasonable Estimates	A mechanism of the WAPC process that is used to adjust the historical volumes used in the formulae if customers are moved to another tariff. The intent of this reasonable estimate mechanism is to minimise gains or losses in revenue of moving customers between tariffs
Second Block	See <i>Inclining Block Tariff (IBT)</i>
Shoulder Consumption Charge	This tariff component applies to the price charged for electricity usage during the shoulder period of the day. See also <i>Time of Day Period Definition</i>
Third Block	See <i>Inclining Block Tariff (IBT)</i>
Time of Day Period Definition	The time periods of the day used to define when the peak, shoulder and off-peak price apply. The following is the current Time of Day Period Definitions: ➤ For Residential and Small Business up to 40 MWh per annum: ▪ Peak: 14:00 – 20:00 working weekdays ▪ Shoulder: 07:00 – 14:00, 20:00-22:00 working weekdays, 07:00 – 22:00 weekends and public holidays ▪ Off Peak: All other times ➤ For all other ToU tariffs ▪ Peak: 14:00 – 20:00 working weekdays ▪ Shoulder: 07:00 – 14:00, 20:00-22:00 working weekdays ▪ Off Peak: All other times
Time of Use (TOU) Tariff	This tariff typically has a fixed charge and a variable energy charge component with different prices applying at different time periods of the day. See also <i>Time of Day Period Definition</i>
Transmission Use Of System (TUOS)	The revenue earned from customers to recover the cost of transmission service provision by TransGrid and other TNSPs
Weighted Average Price Cap (WAPC)	A form of regulation that limits the annual increase to weighted average DUOS price to CPI+X. The weight used in the calculation of the average price is the previous year's volumes
X factor	The allowed rate of growth (positive or negative) in prices (under a price cap) or revenue (under a revenue cap)

Ausgrid FY13 Network Pricing Proposal

Part 2 Demonstrating Compliance



1. Overview and Outline

This document is Ausgrid's formal Pricing Proposal for FY14 and is submitted for review and approval by the Australian Energy Regulator (AER) as required by clause 6.18.2(a)(2) of the Transitional Chapter 6 Rules. It is structured to allow ready assessment of compliance by the AER.

Outline of Compliance with Rules

Ausgrid's Pricing Proposal assesses all of the requirements set out in Rule 6.18.2 included in Chapter 11 (Transitional Rules) of the National Electricity Rules. Our pricing proposal also demonstrates compliance against the applicable distribution determination, as required under Clause 6.18.2(b)(7) of the Transitional Chapter 6 Rules.

Specifically, the proposal sets out the tariff classes, tariffs and charging parameters, and expected revenue for the year commencing 1 July 2013 and ending 30 June 2014.

This pricing proposal is structured as follows:

- Chapter 2 sets out the proposed tariff classes for standard control services, the basis for the proposed tariff classes, and Ausgrid's procedures for the assignment and reassignment of customers to tariff classes;
- Chapter 3 sets out the proposed tariffs and charging parameters, and the matters that Ausgrid has taken into account when determining these tariffs and charging parameters including the long run marginal costs of providing services to each tariff class, the transaction cost implications and the scope for customers to respond price signals;

- Chapter 4 sets out the weighted average revenue by tariff class and compares it to the efficient revenue boundary defined by the standalone cost and avoidable cost;
- Chapter 5 briefly discusses the variations to tariffs;
- Chapter 6 sets out the obligations on Ausgrid to provide revenue under the Climate Change Fund;
- Chapter 7 sets out the approach to passing through the cost of transmission use of system services;
- Chapter 8 sets out the changes since the previous regulatory year, as provided for in the Determination and the Rules; and provides information on how the pricing proposal complies with the requirements for pricing proposals as set out in the Rules.
- Chapter 9 sets out the tariffs and tariff class for Ausgrid's public lighting services for FY14.
- Chapter 10 sets out the proposed new network tariffs for FY14.
- Chapter 11 sets out the proposed reasonable estimates and provides information on how these estimates comply with Appendix J of the Determination.

Ausgrid notes that Attachment D to this proposal demonstrates:

- Compliance with the WAPC control mechanism and side constraint formula
- Compliance with the application of reasonable estimates.
- Compliance with the control mechanism for Ausgrid's prescribed (transmission) standard control services.
- Compliance with the AER's requirements for reporting recovery of TUOS charges to account for under and over recovery of those charges.

2. Network Tariff Classes

RULE REQUIREMENT

Clause 6.18.2(b)(1) of the NER requires that a pricing proposal must set out the tariff classes that are to apply for the relevant regulatory year

In addition, when assigning customers to tariff classes the AER is required to have regard to the following principles;

(1) customers should be assigned to tariff classes on the basis of one or more of the following factors:

(i) the nature and extent of their usage;

(ii) the nature of their connection to the network;

(iii) whether remotely-read interval metering or other similar metering technology has been installed at the customer's premises as a result of a regulatory obligation or requirement;

(2) customers with a similar connection and usage profile should be treated on an equal basis;

(3) however, customers with micro-generation facilities should be treated no less favorably than customers without such facilities but with a similar load profile.

Clause 6.18.3(d) requires that a tariff class be constituted with regard to the need to group customers together on an economically efficient basis, and the need to avoid unnecessary transactions costs.

or sub-transmission voltage (33kV, 66kV and 132kV);

- the usage of the customer (ie, maximum demand exceeding 10MW or total usage of 40 GWh per annum); and
- the metering installed in the customer's premise (ie metered or unmetered).

Table 1 defines Ausgrid's proposed tariff classes for direct control services in FY14.⁸

Definition of a Network Tariff Classes

Under Chapter 10 of the NER, tariff classes are defined as representing 'a class of customers for one or more direct control services, who are subject to a particular tariff or particular tariffs'.

Proposed Network Tariff Classes

Ausgrid proposes to make no changes to the tariff classes for FY14. Customers are assigned to tariff classes on the following basis:

- the voltage connection required by the customer, namely low voltage (415V and 240V), high voltage (5kV, 11kV and 22kV)

⁸ Please note that the tariff classes for Miscellaneous & Monopoly Services are set out in Attachment E. Tariff classes for public lighting services are set out in Chapter 9.

Table 1: Ausgrid's Proposed Tariff Class Descriptions - FY14

Tariff Class	Network Tariff	Definition
Low Voltage	EA010 – Residential Inclining Block EA025– Residential Time of Use EA030 – Controlled load 1 EA040 – Controlled load 2 EA050– Small Business Inclining Block EA225– Small Business Time of Use EA302 – LV 40-160 MWh (System) EA305– LV 160-750 MWh (System) EA310 – LV > 750 MWh (System) EA325 – LV Connection (Standby - Closed)	Applicable to separately metered low voltage (415V or 240V) connections.
High Voltage	EA360 – HV Connection (Standby - Closed) EA370 HV Connection (System) EA380 HV Connection (Substation)	Applicable to any connection at high voltage (11kV) that is not otherwise eligible for a CRNP as defined below.
Sub-transmission	EA390 ST Connection	Applicable to any connection at a sub-transmission voltage (132/66/33kV) that is not otherwise eligible for a CRNP price, as defined below.
Cost Reflective Network Prices (CRNP)	Customer Specific Prices	Applicable to connections at any voltage that use more than 10 MW of electricity demand on at least three occasions over a 12 month period or consume more than 40 GWh over a 12 month period.
Unmetered	EA401 Public Lighting EA402 Constant Unmetered EA403 EnergyLight	Applicable to any LV connection that is defined as an unmetered supply by Ausgrid in consultation with AEMO as per clause S7.2.3(Item 5) of the NER.

Procedures for the assignment and re-assignment of customers to tariff classes

Ausgrid will assign customers to tariff classes by applying the following criteria:

1. A customer's site has a demand in excess of 10 MW on more than three occasions over a consecutive 12 month period, or uses more than 40 GWh per annum, then regardless of the connection voltage, the customer is assigned to the CRNP tariff class for the purpose of demonstrating compliance with the NER at each annual price reset.
2. If the voltage of the supply is 33kV, 66kV or 132kV then the customer is assigned to the Subtransmission Voltage Capacity tariff class (unless already assigned to the CRNP tariff class).
3. If the voltage of the supply is 11 kV then the customer is assigned to the High

Voltage Capacity tariff class (unless already assigned to the CRNP tariff class).

4. If the voltage of the supply is 415V or 240V and the customer's annual consumption, the customer is assigned to Low Voltage tariff class.
5. If the supply is unmetered then the customer is assigned to the Unmetered tariff class.

Ausgrid's criteria for assigning our customers to tariff classes is provided in the following table.

Table 2: Proposed Criteria for Customer Assignment to Tariff Class

Tariff class	Nature of Network Connection	
	Voltage	Metering
Unmetered	415V/240V	Unmetered
Low Voltage ⁹		Metered
High Voltage	5kV, 11kV, 22kV	
Sub-transmission Voltage	33kV, 66kV, 132kV	
CRNP (> 10 MW or 40 GWh)	Any voltage	

⁹ Please note that the low voltage tariff class applies to all residential and business connections connected to the low voltage distribution network, excluding unmetered connections.

If a customer's connection load or characteristics change, the customer will be re-assigned to the appropriate tariff class. This will be carried out annually consistent with the annual review of customers' assignment to network tariffs (as discussed in Attachment G).

The annual review undertaken for this year has identified no need to re-assign existing from one tariff class to another tariff class in FY14.

Customers with micro-generation facilities

It is Ausgrid policy to treat customers with micro-generation facilities no less favourably than customers without these facilities but with a similar consumption profile. Allocation of an embedded generation customer to a tariff class will be made on the same basis as other customers; this being the extent and nature of usage, and the nature of the connection to the network. The network tariff will include fixed and variable components and if the customer's demand were to be met entirely by the micro-generator then the levied charge will only be the fixed connection component.

Ausgrid's compliance with this requirement can be demonstrated by the proposed network pricing arrangements for customers participating in the NSW Government Solar

Bonus Scheme, which ensures that these customers are treated no less favourably than other customers as the billed consumption of these customers will be unaffected by their participation in the scheme. However, customers will be required to be on a network ToU tariff to participate in the NSW Government Solar Bonus Scheme. Ausgrid does not intend to account for the tariff transfers arising from the solar bonus scheme in reasonable estimates because Appendix J of the Determination specifically excludes voluntary transfers.

3. Proposed Tariffs and Charging Parameters

RULE REQUIREMENT

Clause 6.18.2(b)(2) requires that the pricing proposal set out the proposed tariffs for each tariff class; and

Clause 6.18.2(b)(3) requires that the pricing proposal set out the charging parameters and the elements of service to which each charging parameter relates,

Rule 6.18.5 sets out the pricing principles that are relevant when determining tariffs and charging parameters.

Rule 6.18.5 further provides that:

(b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class:

(1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and

(2) must be determined having regard to:

(i) transaction costs associated with the tariff or each charging parameter; and

(ii) whether customers of the relevant tariff class are able or likely to respond to price signals.

(c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

This chapter sets out the proposed tariffs and charging parameters for direct control services for each tariff class for the period of 1 July 2013 to 30 June 2014, and explains the elements of service to which each charging parameter relates. The charging parameters for alternative control services (public lighting services) are discussed in Chapter 9 and set out in Attachments B and C. The charging parameters for Miscellaneous and Monopoly Services are in Attachment E.

The proposed tariffs and charging parameters are set out for each tariff class in Table 3A, 3B 3C and 3D. This chapter also explains how Ausgrid has applied the pricing principles set out in clause 6.18.5 of the NER. In developing the charging parameters, Ausgrid has taken account of a range of estimates of long run marginal cost, and has had regard to transactions costs and the scope for customers to respond to price signals. In addition, Ausgrid has ensured that its network tariffs are designed to recover the difference between marginal and average cost with the minimum of distortion to efficient patterns of consumption.

Table 3a: Ausgrid's Proposed FY14 Network Tariffs by Charging Parameter (Exclusive of GST) - DUOS

Tariff Class	Tariff Code	Tariff Name	Network Access Charge c/day	Network Energy Prices					Daily Capacity Prices			
				Non-TOU c/kWh	Peak c/kWh	Shoulder c/kWh	Off-peak c/kWh	Block 1 c/kWh	Block 2 c/kWh	Block 3 c/kWh	Peak c/kW/day	Peak c/kVA/day
Low Voltage	EA010	LV Res non-TOU	38.9500					12.2134	1.2093			
	EA025	LV Res < 40 MWh (System)	51.2500		14.7399	4.2927	2.1744					
	EA030	Controlled load 1	3.2390	0.0105								
	EA040	Controlled load 2	12.5255	0.0260								
	EA050	LV Business non-TOU(Closed)	123.0000					9.9926	4.8309			
	EA225	LV Business TOU	123.0000		10.3271	4.6543	0.8281					
	EA302	LV 40-160 MWh (System)	563.7500		13.1207	5.4449	2.8864				33.8250	
	EA305	LV 160-750 MWh (System)	1,640.0000		10.0490	4.9281	1.8584					33.8250
	EA310	LV > 750 MWh (System)	2,048.9750		10.1202	4.8476	1.7779					33.8250
	EA325	LV Connection (Closed)	2,048.9750		7.3159	5.4207	1.7365					
High Voltage	EA360	HV Connection (Closed)	1,845.0000		4.9698	2.5206	0.2194					
	EA370	HV Connection (System)	4,100.0000		4.8001	1.2157	0.5962					17.7325
	EA380	HV Connection (Substation)	4,100.0000		3.1400	1.6571	0.1171					17.7325
Sub-transmission	EA390	ST Connection	5,125.0000		0.0235	0.0166	0.0098					5.6785
Unmetered	EA401	Public lighting		6.4409								
	EA402	Constant unmetered		7.8616								
	EA403	EnergyLight		5.3449								

Table 3b: Ausgrid's Proposed FY14 Network Tariffs by Charging Parameter (Exclusive of GST) - TUOS

Tariff Class	Tariff Code	Tariff Name	Network Access Charge c/day	Network Energy Prices					Daily Capacity Prices			
				Non-TOU c/kWh	Peak c/kWh	Shoulder c/kWh	Off-peak c/kWh	Block 1 c/kWh	Block 2 c/kWh	Block 3 c/kWh	Peak c/kW/day	Peak c/kVA/day
Low Voltage	EA010	LV Res non-TOU						0.2370	13.7011	19.0104		
	EA025	LV Res < 40 MWh (System)			11.1295	0.5846	0.1815					
	EA030	Controlled load 1		1.8841								
	EA040	Controlled load 2		5.0461								
	EA050	LV Business non-TOU(Closed)						0.5724	14.4466			
	EA225	LV Business TOU			10.0128	1.1191	0.4467					
	EA302	LV 40-160 MWh (System)			0.1770	0.1653	0.1613					
	EA305	LV 160-750 MWh (System)			0.2352	0.2311	0.2257					
	EA310	LV > 750 MWh (System)			0.3770	0.3196	0.3143					
	EA325	LV Connection (Closed)			0.3716	0.3193	0.3135					
High Voltage	EA360	HV Connection (Closed)			3.0560	2.2251	1.5436					
	EA370	HV Connection (System)			3.0629	3.3673	1.0040					
	EA380	HV Connection (Substation)			3.5647	2.8950	0.8681					
Sub-transmission	EA390	ST Connection			4.0161	2.5881	1.2111					
Unmetered	EA401	Public lighting		1.6772								
	EA402	Constant unmetered		2.2347								
	EA403	EnergyLight		1.9531								

Table 3c: Ausgrid's Proposed FY14 Network Tariffs by Charging Parameter (Exclusive of GST) - CCF

Tariff Class	Tariff Code	Tariff Name	Network Access Charge c/day	Network Energy Prices					Daily Capacity Prices			
				Non-TOU c/kWh	Peak c/kWh	Shoulder c/kWh	Off-peak c/kWh	Block 1 c/kWh	Block 2 c/kWh	Block 3 c/kWh	Peak c/kW/day	Peak c/kVA/day
Low Voltage	EA010	LV Res non-TOU						0.4646	0.4646	0.4646		
	EA025	LV Res < 40 MWh (System)			0.2681	0.2681	0.2681					
	EA030	Controlled load 1		0.2560								
	EA040	Controlled load 2		0.2560								
	EA050	LV Business non-TOU(Closed)						0.9560	0.9560			
	EA225	LV Business TOU			0.7751	0.7751	0.7751					
	EA302	LV 40-160 MWh (System)			0.8268	0.8268	0.8268					
	EA305	LV 160-750 MWh (System)			0.8268	0.8268	0.8268					
	EA310	LV > 750 MWh (System)			0.7881	0.7881	0.7881					
	EA325	LV Connection (Closed)										
High Voltage	EA360	HV Connection (Closed)										
	EA370	HV Connection (System)			0.7881	0.7881	0.7881					
	EA380	HV Connection (Substation)			0.7881	0.7881	0.7881					
Sub-transmission	EA390	ST Connection			0.7881	0.7881	0.7881					
Unmetered	EA401	Public lighting		0.7934								
	EA402	Constant unmetered		0.7934								
	EA403	EnergyLight		0.7934								

Table 3d: Ausgrid's Proposed FY14 Network Tariffs by Charging Parameter (Exclusive of GST) - NUOS

Tariff Class	Tariff Code	Tariff Name	Network Access Charge c/day	Network Energy Prices				Daily Capacity Prices				
				Non-TOU c/kWh	Peak c/kWh	Shoulder c/kWh	Off-peak c/kWh	Block 1 c/kWh	Block 2 c/kWh	Block 3 c/kWh	Peak c/kW/day	Peak c/kVA/day
Low Voltage	EA010	LV Res non-TOU	38.9500					12.9150	15.3750	19.4750		
	EA025	LV Res < 40 MWh (System)	51.2500		26.1375	5.1455	2.6240					
	EA030	Controlled load 1	3.2390	2.1506								
	EA040	Controlled load 2	12.5255	5.3281								
	EA050	LV Business non-TOU(Closed)	123.0000					11.5210	20.2335			
	EA225	LV Business TOU	123.0000		21.1150	6.5485	2.0500					
	EA302	LV 40-160 MWh (System)	563.7500		14.1245	6.4370	3.8745				33.8250	
	EA305	LV 160-750 MWh (System)	1,640.0000		11.1110	5.9860	2.9110					33.8250
	EA310	LV > 750 MWh (System)	2,048.9750		11.2853	5.9553	2.8803					33.8250
	EA325	LV Connection (Closed)	2,048.9750		7.6875	5.7400	2.0500					
High Voltage	EA360	HV Connection (Closed)	1,845.0000		8.0258	4.7458	1.7630					
	EA370	HV Connection (System)	4,100.0000		8.6510	5.3710	2.3883					17.7325
	EA380	HV Connection (Substation)	4,100.0000		7.4928	5.3403	1.7733					17.7325
Sub-transmission	EA390	ST Connection	5,125.0000		4.8278	3.3928	2.0090					5.6785
Unmetered	EA401	Public lighting		8.9115								
	EA402	Constant unmetered		10.8898								
	EA403	EnergyLight		8.0915								

There are three elements of service for which charging parameters are defined, namely:

- Network access charge per day per connection (c/connection/day) – this is a flat charge per connection, charged on the basis of the number of days of access to the network for that connection. A connection is defined as any connection to the network requiring a unique National Metering Identifier (NMI) as defined by the AEMO National Metering Identifier Procedure;
- Electricity usage charge (c/kWh) – this is a charge per unit of electricity supplied to a connection point, through a primary tariff or also through an additional secondary tariff (usually controlled load hot water as seen from the Non-ToU charging parameter); and
- Capacity charge per kilowatt or kiloVoltAmp, per day (c/kW/day or c/kVA/day) – this is a charge based on a customer's maximum demand recorded on a working weekday, in any half-hour period between 2pm and 8pm over the twelve months preceding the end of the current billing month. Maximum demand is

recorded in either kW or kVA according to the customer's metering functionality.

In addition, there are a number of alternative components for the electricity usage charge including:

- An inclining block charge – this charge which is triggered once a threshold amount of electricity use has been used in each billing quarter. All electricity used after this point is charged at a higher rate based on (¢/kWh). For example, the residential IBT structure has three blocks with annual energy consumption thresholds at 0 to 4000 kWh, 4000 to 8000 kWh and greater than 8000 kWh. The business IBT structure has two blocks with an annual energy consumption threshold at 10,000 kWh pa.
- Time of use consumption charges vary according to the time of day of consumption and whether it is a weekday, or weekend/public holiday, as summarised in Table 4.

Table 4: Time of Use periods for Tariffs

Customer Type	Annual consumption	Time of Use period definition
Residential and Small Business	< 40 MWh per annum	Peak period: 2.00 pm – 8.00 pm on working weekdays. Shoulder period: 7.00 am – 2.00 pm and 8.00 pm – 10.00 pm on working weekdays and from 7.00 am – 10.00 pm on weekends and public holidays Off-Peak period : All other times
Medium and Large Business	> 40 MWh per annum	Peak period: 2.00 pm – 8.00 pm on working weekdays. Shoulder period: 7.00 am – 2.00 pm and 8.00 pm – 10.00 pm on working weekdays Off-Peak period : All other times

Ausgrid uses the Inclining Block Tariff (IBT) structure for customers with Type 6 metering installations. The inclining block pricing structure is designed to encourage energy conservation, particularly for customers with air-conditioners who use significantly more

energy (and contribute more to summer peak network congestion) than other customers. The structure of the IBT also provides a financial incentive for customers with larger energy consumption to switch to a more cost reflective time of use tariff.

The price differential between the first, second and third block consumption under the IBT structure has been set based on an assessment of customers' capacity to respond to the price signal. Currently the second and third block prices are significantly higher (at the NUOS level) than the price charged for first block consumption.

The IBT is designed to provide a strong financial incentive (ie higher network bills) to customers with significant second and third block consumption to change their consumption behaviour. Importantly from an economic perspective, the IBT also provides larger users with the financial incentive to voluntarily switch to the more cost reflective ToU tariff. The following figure demonstrates that the majority of customers and certainly all large energy users will be financially better off from moving to the TOU tariff. This approach is consistent with Ausgrid commitment to pursuing network tariff reform in a socially responsible manner by striking an appropriate balance between economic and equity objectives.

Ausgrid believes that the three block structure introduced in FY 2013 represents the best option (under basic accumulation metering) in terms of ensuring residential IBT customers receive more appropriate incentives to change their consumption behavior or to voluntarily switch over to the ToU tariff.

Ausgrid remains aware that any change to the price differential and consumption thresholds under the IBT structure has the potential to create material adjustment costs for some customers. Therefore the economic benefits of reform will need to be considered in the context of other considerations, such as equity and the preference of customers for simple tariffs. Ausgrid intends to transition the price level under this new structure to give customers more time to adjust their

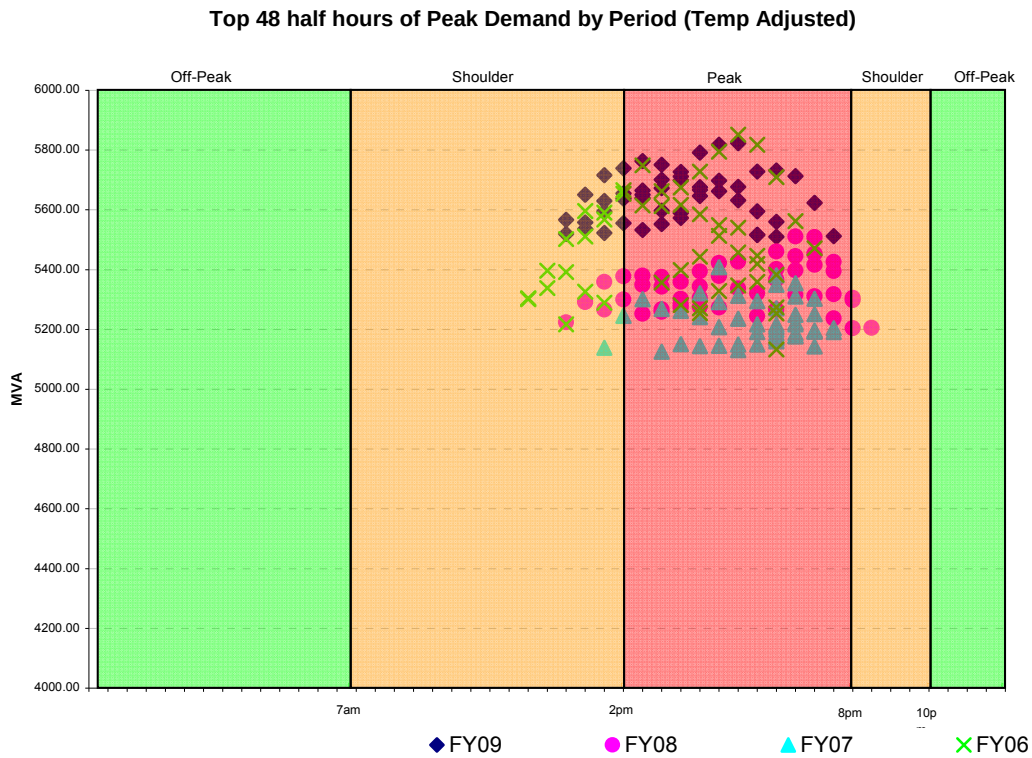
consumption patterns to mitigate any adverse bill impacts arising from the reform process.

There are also opportunities to improve the design of the time of use tariff from an economic perspective. The current period definition for time of use (peak, shoulder and off-peak) is designed to cover both the evening winter and summer peak across the whole network. Ausgrid believes that the broad definition of the peak period which covers all business days throughout the year has the potential to undermine economic efficiency even if peak energy charges are set reflective of LRMC because the network is not congested on all business days. Ausgrid intends to investigate this issue as part of our major review of our tariff strategy prior to the commencement of the next regulatory control period.

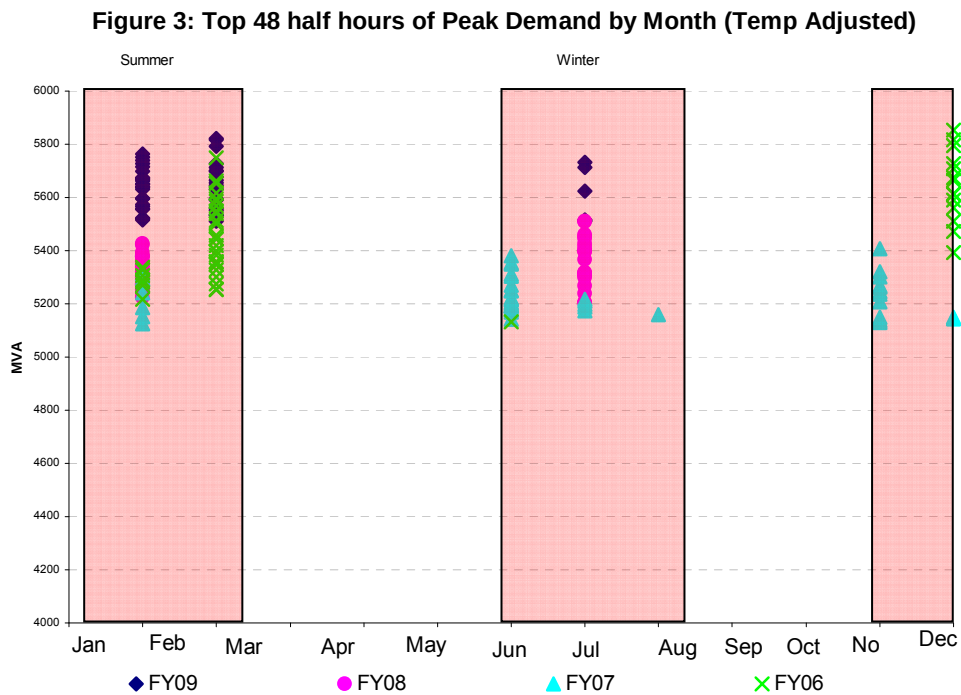
As with the reforms undertaken to the residential IBT structure, Ausgrid will ensure that external stakeholders are informed of developments relating to the ToU tariff structure. Ausgrid's intention for reform is to maximise economic benefits without imposing material adjustment costs on affected customers. Ausgrid believes that this approach to tariff reform, together with a good communication strategy, will encourage permanent changes in consumption behaviour by providing customers (and their retailers) with greater certainty over the long-term direction of prices and pricing structures. It is for this reason that Ausgrid proposes to maintain the current price differences between consumption blocks under the IBT in FY 2014.

An analysis of the timing of the top 48 half hourly peak consumption intervals over recent years indicates that there are likely to be economic benefits to be realised from further reform to the peak period to ensure that peak price signals better reflect emerging patterns of network congestion.

Figure 2: Historical Pattern of Peak Demand – Half Hour Intervals



The following figure shows that the peak demand intervals occur in the warmer and cooler months of the year.



Further analysis indicates that the nature of peak network congestion is different in winter and summer, and is presented in Figure 4 and Figure 5 respectively.

Figure 4: Winter Peak Demand by Time Period (Temp Adjusted)

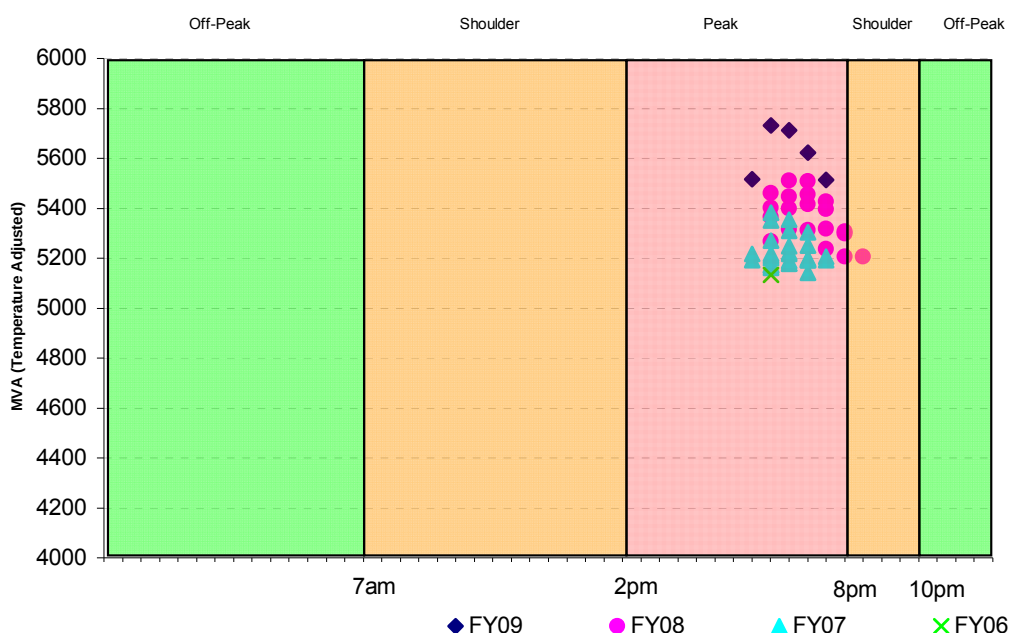
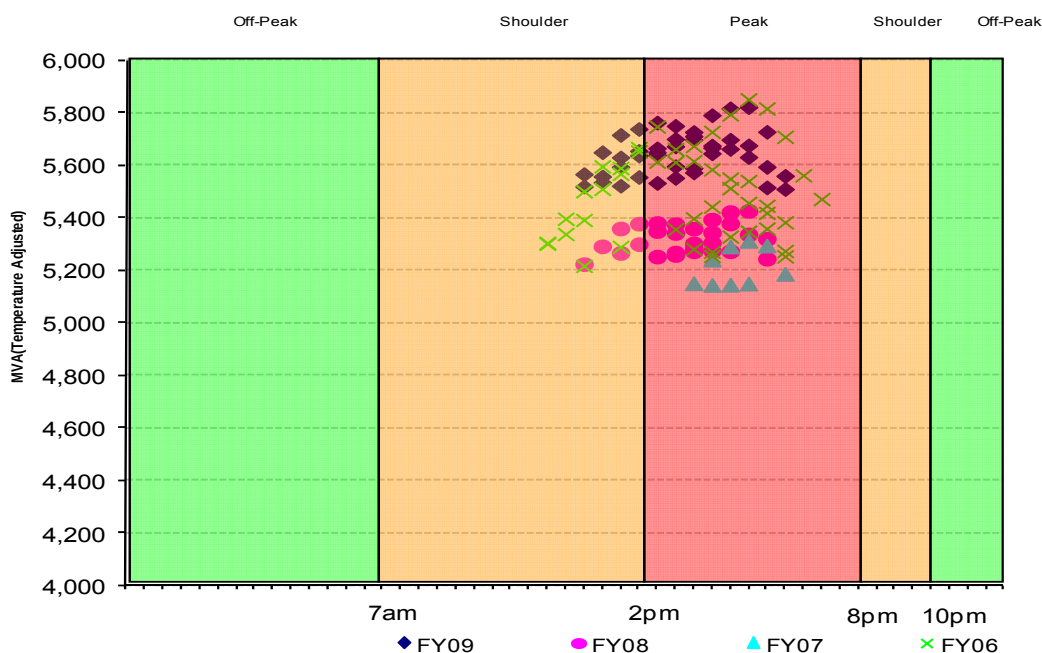


Figure 5: Summer Peak Demand by Time Period (Temp Adjusted)



Long run marginal costs

The concept of marginal costs is important in economics because of the general principle that setting prices equal to marginal costs presents consumers with the opportunity costs of their consumption decisions, and results in an efficient allocation of society's resources.

The timeframe over which marginal costs are considered to vary with consumption is the distinguishing difference between the concepts of short-run marginal cost (SRMC) and long-run marginal cost (LRMC). In the short-run only costs that directly vary with demand can be avoided (such as the need for some repairs and maintenance, or call centre costs), while in

the long-run capital costs can be avoided particularly if anticipated demand growth does not eventuate. The LRMC is equal to the SRMC plus the marginal capital costs.

The use of LRMC as a basis for determining usage charges is common in network businesses to provide customers with signals about the incremental capital and operating costs associated with use of the network. The use of LRMC is preferred over SRMC because of a desire to minimise the scope for network constraints to ration use of the network. In principle, the use of LRMC ensures that sufficient network capacity is made available such that demand is always able to be met by supply capacity.

Importantly, LRMC is a forward looking concept and should be estimated taking into account future expectations of the growth in costs to meet expected increases in demand. There are a number of methods commonly used to estimate the LRMC for network businesses including the Average Incremental Cost (AIC) approach and the Turvey methodology.

These two approaches differ in the methodology used to estimate the incremental impact of changes in demand on capacity related expenditure. The AIC approach calculates the average of expected capacity related expenditure over the period that output is expected to change. In contrast the Turvey approach directly estimates the change in expenditure resulting from an increment or decrement in forecast demand.

While there are strong theoretical underpinnings in support of the Turvey approach, Ausgrid has instead estimated the LRMC for each of the proposed tariff classes using the AIC approach. That said it intends to investigate the appropriateness of alternative approaches to estimating LRMC for tariff classes for future pricing proposals.

The AIC approach can be expressed algebraically as follows:

$$LRMC (AIC) = \frac{PV(capex) + PV(opex)}{PV(incremental_demand)}$$

where the optimal capital costs are the annualised capital expenditure required to meet the additional demand forecast over the period and the operating costs are those required to serve additional demand over the forecast period. The additional demand served is the forecast change in demand compared to a base year. Ausgrid considers that incremental demand (rather than energy

consumption) is the most appropriate volume cost driver for the LRMC analysis.

The Determination sets out Ausgrid's required capital expenditure program to meet its overall demand forecasts. This information has been used as the basis for determining the optimal capital costs to be allocated to each individual network tariff. Similarly the additional demand served has also been allocated to these network tariffs. The resulting estimates of the LRMC were aggregated by proposed network tariff class as set out in the following table.

Table 5: Ausgrid's Long Run Marginal Cost

Tariff Class	LRMC Estimate (\$/kVA)
Low Voltage ¹⁰	\$152.30
High Voltage	\$166.25
Sub-transmission Voltage	\$38.04
CRNP	\$28.14
Unmetered	\$75.44

Ausgrid LRMC model is extremely sensitive to key assumptions and inputs. Ausgrid believes that there is further scope to improve its estimates of LRMC and remains committed to undertaking a major review of recent economic literature in this area of the price setting function prior to the commencement of next regulatory control period. It is for this reason that Ausgrid proposes to update the LRMC estimates for FY 2014 to reflect only movements in CPI. A more robust LRMC estimate will be provided to external stakeholders once the review has been completed and the findings of this review have been fully incorporated into our modeling process.

The application of LRMC to the setting of efficient network tariffs is a two step process, as summarised below:

- To select the charging parameter to use to signal LRMC to customers.

¹⁰ Please note that the low voltage tariff class applies to all residential and business connections connected to the low voltage distribution network, excluding unmetered connections.

- To convert the LRMC estimate from a \$ per kVA basis to a cents per kWh basis, if required.

Ausgrid believes it is efficient (and appropriate) that LRMC estimates be reflected only in the setting of charging parameter(s) that cover the period when there is a material likelihood of system-wide network congestion, which are a key driver of network augmentation costs. A review of Ausgrid's existing charging parameters indicates that there are strong economic grounds to reflecting LRMC in the setting of peak energy charges given that there is a reasonable probability that usage during peak periods will be correlated with network driving augmentation costs.¹¹ It is important to note that future network tariff reforms could result in the capacity charge, rather than peak energy charges, being the most efficient charging parameter to reflect LRMC. The potential reform options for the capacity charging parameter will focus on improving the economic efficiency of the peak price signal by recognising that marginal costs of peak network capacity vary by the following dimensions:

- Time of year – the network is generally constrained during high peak demand periods of summer and winter;
- Time of day – peak demand during the day tends to reflect temperature conditions and their impact on heating and cooling loads; and
- Location in the network – the balance between peak demand and network capacity tends to vary within the network area.

In recognition of the importance of LRMC based pricing to customers and other external stakeholders, Ausgrid intends to review the methodology and modelling of LRMC in the near future to ensure that this important basis to the setting of peak charges is consistent with best practice. As with other areas of potential reform, Ausgrid will ensure that external stakeholders are informed of any changes to the calculation of LRMC over the remainder of the current regulatory period and the methodological and/or modelling basis of these changes.

Given the decision to reflect LRMC in the peak energy charges, it is necessary to convert the

LRMC estimates for each tariff class from a \$ per kVA basis to a cents per kWh basis. The formula used by Ausgrid to make this conversion is shown below:

$$LRMC = \frac{LRMC\text{ Estimate}(\$/kVA)}{\text{peak hours per annum} \times \text{power factor}}$$

Applying the above formula results in the conversion of the LRMC estimate from a \$ per kVA to a cents per kWh basis, at each voltage level, as summarised in the table below:

¹¹ This correlation is likely to strengthen if the peak period definition is reformed to more accurately reflect summer and winter patterns of network congestion.

Table 6: Ausgrid's Long Run Marginal Cost Estimate

Tariff Class	Annual Hours of Peak Period	Assumed Power Factor	Range of LRMC Estimate Cents per KWh		
			Lower Bound	Point Estimate	Upper Bound
Low Voltage	1506	0.85	6.7	11.0	25.0
High Voltage	1506	0.90	3.6	6.0	16.3
Sub-transmission Voltage	1506	0.95	1.7	2.8	7.7
CRNP	1506	0.95	2.4	5.1	∞
Unmetered	1506	0.85	7.5	12.8	35.0

As required by the NER, Ausgrid has taken into account the estimated range of LRMC shown in the above table when setting the peak energy charging parameter of network tariffs for FY14. To achieve an efficient pricing outcome, the peak charge in DUOS terms has been set consistent with this LRMC range in FY14. This has meant that for some network tariffs, where the current peak charge is above or close to the upper range of LRMC, the increase to the peak charge has been minimised in DUOS terms in FY14. This has resulted in a larger DUOS increase being applied to the other charging parameters than would otherwise have been the case. As required under clause 6.18.5(c), this rebalancing of DUOS tariffs has been undertaken in a manner that minimises the distortion to efficient consumption patterns.

It is also important to note that Ausgrid has undertaken considerable rebalancing of TUOS tariffs in FY14 to ensure that the pursuit of more efficient DUOS pricing does not create unacceptable NUOS price trend uncertainty for network customers. Ausgrid strongly believes that this is a reasonable approach to adopt considering the need to balance economic objectives with the need to provide customers with a reasonable degree of pricing certainty at the NUOS level.

Transactions costs

In developing the tariffs and charging parameters, Ausgrid has been mindful of minimising the transactions costs associated with levying charges. The decision to charge on the basis of a combination of fixed, usage

and capacity charges reflects the trade off between applying a complex set of charges (eg, introducing a power factor charge or additional inclining block rates) and ensuring that charges can be closely aligned to the underlying marginal costs of providing direct control services.

Introducing a greater number of tariff and charging parameters increases pricing complexity and will inevitably increase the transactions costs involved in charging for network services, both to Ausgrid through the need for more complex (and expensive) customer management and billing systems, and to customers, through additional effort required to understand the basis upon which charges are calculated and determine the appropriate response.

Ausgrid believes that its network tariffs and charging parameters strike the right balance between transactions costs and ensuring that appropriate signals are provided to customers to facilitate the efficient use of network services.

Scope to respond to price signals

Ausgrid's network tariffs and charging parameters provide both short term and long term incentives to customers to modify consumption patterns in line with efficient use of network services. These price signals are conveyed to consumers of network services through the use of energy and/or capacity charges, depending on metering constraints, and have been designed with the principal aim of ensuring that network customers make efficient decisions about their use of the

distribution and transmission network. To the extent that retailers pass through Ausgrid's network price signal, it is possible that end-customers may respond to these efficient price signals by changing their consumption behaviour by moving electricity usage out of the time-of-use peak periods. Over the long term efficient prices could encourage the installation of more effective insulation for housing, more energy efficient plant and equipment, or the installation of power factor correction devices e.g. capacitor banks.

Ausgrid has undertaken a trial to test residential and business customer demand response to new tariff initiatives. The Strategic Pricing Study (SPS) was a two year study that was completed in August 2008, and has been reviewed in previous pricing proposals. The results of the trial were encouraging. Over the 14 events that were called during the two year period, residential customers showed energy reductions of up to 30% (winter) and 36% (summer), when compared to the control group.

Ausgrid is continuing to explore the potential of innovative tariffs during the current regulatory period with our involvement in the Smart Grid Smart City project (refer to Chapter 5 of Part 1 of this document).

4. Weighted Average Revenue

RULE REQUIREMENT

Clause 6.18.2(b)(4) of the NER requires that a pricing proposal set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year.

it compares these revenues with the standalone costs and avoidable costs by tariff class to demonstrate that the proposed tariffs comply with the principles set out in the NER.

The weighted average revenue for FY13 and FY14 is set out in the following table.

This chapter sets out the expected weighted average revenue from tariffs within each tariff class for standard control services. In addition,

Table 7: Expected DUOS Revenue by Tariff Class (\$ million excluding GST)

Tariff Class	Weighted Average Revenue	
	FY13(\$m)	FY14(\$m)
Low Voltage	2,047.31	1,982.93
High Voltage	49.29	44.11
Sub-transmission Voltage	6.05	5.28
Unmetered	14.14	13.23
CRNP	49.47	50.20

Estimating standalone costs

The standalone costs represent the costs that would be incurred to replicate or bypass the infrastructure used to provide a service. It represents an upper bound of costs that should be recovered from customers in a particular tariff class. If customers were paying above the standalone costs then it follows that it would be economically beneficial for customers to switch to an alternative provider, and similarly if entry was economically feasible it would result in new suppliers entering to provide services. This creates the possibility of inefficient bypass of the infrastructure.

To estimate the standalone costs for a tariff class, Ausgrid has used the construct of a hypothetical new entrant distribution network service provider that is seeking to supply

services to each tariff class separately. Ausgrid has then asked what infrastructure and operating costs would be incurred to supply the tariff class alone, without the benefit of any economies of scale and scope arising from the supply of services to all other tariff classes. To assist with estimating the cost of the hypothetical new entrant, Ausgrid has drawn upon information from its own costs as follows:

- Identification of the segments of the network that would be required to provide services to each tariff class. For example, for HV Business customers we identified the cost of replicating the network from the point of customer connection to the high voltage network to the connection with the TransGrid transmission network.

- Develop an estimate of the annual capital charge for the relevant network by applying the same return to capital and depreciation parameters as set out in the Determination.
- Develop an estimate of the operating costs that would be incurred in order to provide services to the tariff class by identifying those operating costs that are necessarily incurred by Ausgrid for the provision of network services to that tariff class.

The associated summation of the annual estimated capital and operating costs for each tariff class are therefore estimated to represent the standalone costs that would be incurred by a hypothetical new entrant.

Ausgrid believes that there is scope to improve its estimation of standalone cost and intends to undertake a review prior to the commencement of the next regulatory control period. It is for this reason that Ausgrid proposes to update the standalone estimates for FY 2014 to reflect only movements in CPI. A more robust standalone cost estimate will be provided to external stakeholders once the review has been completed and the findings of this review have been fully incorporated into our modeling process.

The estimated standalone costs for each tariff class are set out in the following table.

Table 8: Comparison of Standalone Costs Vs FY14 DUOS Tariffs

Tariff Class	Total Standalone Cost	Weighted Average Revenue	Weighted Average Revenue as a proportion of Standalone Cost
	FY14 (\$m)	FY14 (\$m)	FY14 (%)
Low Voltage	2,166.23	1,982.93	92%
High Voltage	567.98	44.11	8%
Sub-transmission Voltage	206.16	5.28	3%
Unmetered	1,294.97	13.23	1%
CRNP	572.98	50.20	9%

It can be seen that costs need to establish and maintain a standalone service for each tariff class is higher than the weighted average revenue to be recovered in FY14.

Estimating avoidable costs

The avoidable costs represent those costs that could be avoided by a business if it was not supplying goods or services to its customers. It represents the lower bound of costs that should be recovered from customers. If customers were charged below the avoidable costs it would be economically beneficial for the business to stop supplying the customer and thereby avoid the associated costs which would exceed the revenue expected to be obtained from the customer.

To estimate the avoidable costs for each proposed tariff class, Ausgrid has:

- identified those categories of costs that would be avoided if a tariff class was no longer served;
- allocated the cost of each avoidable cost category to each tariff class based on either volume or customer numbers, according to the nature of the cost category, to determine those costs that would be reduced if a particular tariff class was no longer served;
- summed the allocated avoidable cost for all avoidable cost categories for each tariff class to estimate the total avoidable cost for each tariff class.

The cost categories that were identified as being avoidable included:

- repairs and maintenance – this was identified as a partially avoidable cost given that use of the network impacts on its deterioration and the need for repairs and maintenance;
- customer service – these costs relate to managing billing and customer service enquiries which are related to the number of customers served;
- metering costs – these costs relate to meter reading and meter replacement and are related to the number of customers served;
- corporate and divisional support costs – costs relating to media, marketing and legal expenses can be considered to be driven by the number of customers served; and
- customer connections and installation inspections – these costs can be considered to be proportional to customers

numbers, the voltage and capacity of the connection.

Ausgrid believes that there is scope to improve its estimation of avoidable cost and intends to undertake a review prior to the commencement of the next regulatory control period. It is for this reason that Ausgrid proposes to update the avoidable cost estimates for FY 2014 to reflect only movements in CPI. A more robust avoidable cost estimate will be provided to external stakeholders once the review has been completed and the findings of this review have been fully incorporated into our modeling process.

The following table sets out the avoidable costs for each of the tariff classes.

Table 9: Comparison of Avoidable Costs Vs FY14 DUOS Tariffs

Tariff Class	Total Avoidable Costs	Weighted Average Revenue	Avoidable Cost a proportion of expected revenue
	FY14 (\$m)	FY14 (\$m)	FY14(%)
Low Voltage	267.64	1,982.93	13%
High Voltage	5.77	44.11	13%
Sub-transmission Voltage	3.64	5.28	69%
Unmetered	0.89	13.23	7%
CRNP	8.46	50.20	17%

It can be seen that in most cases the distribution revenue to be recovered for each tariff class is well above the avoidable cost for each tariff class. The avoidable cost accounts for a higher share of weighted average revenue for the sub-transmission and CRNP tariff classes, reflecting the greater influence of customer-specific network assets.

As required under clause 6.18.2 of the NER, the following table provides a comparison of

weighted average distribution revenue in FY14 with the estimates of standalone and avoidable costs for each tariff class. As required under the NER, the weighted average distribution revenue lies within the bounds of the upper limit (standalone cost) and the lower limit (avoidable cost)

Table 10: Efficient Pricing Bounds Test

Tariff Class	Total Avoidable Costs	Weighted Average Revenue	Total Standalone Cost
	FY14 (\$m)	FY14 (\$m)	FY14 (\$m)
Low Voltage	267.64	1,982.93	2,166.23
High Voltage	5.77	44.11	567.98
Sub-transmission Voltage	3.64	5.28	206.16
Unmetered	0.89	13.23	1,294.97
CRNP	8.46	50.20	572.98

Recovery of Total Regulated Revenues

This section briefly describes the WAPC formula and the approach adopted to estimate consumption to be used in the formula when new tariffs are introduced, or customers move between tariffs.

The Weighted Average Price Cap (WAPC) formula governing Ausgrid's DUOS pricing is defined below:

$$\frac{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^t * q_{ij}^{t-2}}{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{t-1} * q_{ij}^{t-2}} \leq (1 + \Delta CPI) * (1 - X_t) * (1 + D_t)$$

$i=1, \dots, n$ and $j=1, \dots, m$

where:

n is the number of distribution tariffs

m is the number of distribution tariff components

p_{ij}^t is the proposed price for component j of the distribution tariff i for Year t

p_{ij}^{t-1} is the price charged by the DNSP for aggregate component j of the distribution tariff i in Year $t-1$ (ie the Year immediately preceding Year t)

q_{ij}^{t-2} is the audited quantity of aggregate component j of the distribution tariff i charged by the DNSP in Year $t-2$ (ie the Year immediately prior to Year $t-1$)

X_t is the value of "X" specified for the DNSP (for the Year t) by the AER

ΔCPI is the December to December change in the Consumer Price Index from Year $t-2$ to Year $t-1$

An important feature of the WAPC formula is the use of lagged electricity consumption data to demonstrate compliance. Lagged data is used because of the delays in this data being audited and made available, and the uncertainty associated with the use of forecast volume data within the formula.

The use of lagged consumption data impacts on Ausgrid's ability to recover its revenue requirement in circumstances when actual consumption in a year varies considerably from lagged consumption. This can occur when new tariffs or charging parameters are introduced, or when customers move between tariffs.

In these circumstances, Ausgrid applies a 'reasonable estimates' methodology to calculate the volume to be included in the WAPC formula. This ensures that revenue is maintained despite the tariff changes. For each year of the 2009 to 2014 distribution determination, Ausgrid will submit reasonable estimates of consumption to the AER for use in the WAPC formula if appropriate.

These estimates are based on the quantities that would have been sold in year $t-2$ if the new tariffs or tariff components had been introduced in that same year, or if Ausgrid mandated the transfer of customers from one tariff to another. Voluntary tariff transfers are not included in reasonable estimates.

Finally, please note that Ausgrid proposes to reduce DUOS prices on average under the WAPC by around 0.9% to ensure that our published tariff customers receive an increase at the NUOS level in FY14 close to CPI. This

proposal is expected to result in our DUOS revenue under the WAPC being around \$36m lower than if DUOS prices under the WAPC had increase by the allowed 1.08% in FY14.

5. Variations to Tariffs

RULE REQUIREMENT

Clause 6.18.2(b)(5) of the NER requires that a pricing proposal set out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur

Ausgrid does not propose to make any variation or adjustment to the proposed published and CRNP tariffs during the course of FY2014.

Ausgrid also proposes to introduce no new published tariffs in FY 2014. However, please note that our pricing proposal for FY14 includes some draft pricing trial network tariffs. These trial tariffs relate to the Smart City Smart Grid initiative and a potential Ausgrid demand management trial currently under investigation. It is possible that Ausgrid will need to vary or adjust some of these tariffs to ensure that research objectives are met. Given the voluntary nature of the pricing trial Ausgrid has made no reasonable estimate of the quantities assigned to these tariffs has been made for period t.¹² Given the uncertainties surrounding the take-up, opt-out and the energy consumption response of customers on these trials, Ausgrid has made no attempt to forecast the volumes associated with these tariffs in FY14.¹³

¹² Reasonable estimates are not permitted for voluntary tariff re-assignments, as per Appendix J of the Determination.

¹³ Excluded from the forecast due to uncertainty over the voluntarily take-up rate for these new trial tariffs.

6. Climate Change Fund

RULE REQUIREMENT

Clause 6.18.2(b)(5A) of the NER requires that a pricing proposal must set out the amount paid to the Climate Change Fund in or in respect of the relevant regulatory year and any adjustments to tariffs resulting from an over or under recovery of these amounts in any previous regulatory year. The proposal must also show the amount attributed to the recovery of the Climate Change Fund in terms of expected revenue for the relevant regulatory year

For the year ending June 2014, Ausgrid will pay \$149.8M to the Climate Change Fund. Ausgrid expects to recover \$149.7M for this purpose from network charges, as shown in the following table.

Attachment F is the correspondence received from the Department of Environment and Climate Change on Ausgrid's contribution to the Climate Change Fund for FY14.

Table 11: Overs and Unders Account Forecast Closing Balance – Climate Change Fund

Financial Year Ending	Units	Period t-2	Period t-1	Period t
		Expected FY12	Forecast FY13	Forecast FY14
Interest rate applicable to balance	%	10.02	10.02	10.02
Opening balance	\$'000	(313)	(10)	94
Interest on opening balance (365 days)	\$'000	(31)	(1)	9
Forecast over/under recover for financial year	\$'000	319	100	(99)
Interest charged on over/under recovery for financial year	\$'000	16	5	(5)
Closing balance of CCF overs/unders account	\$'000	(10)	94	0.0

7. Transmission Use of System Services

RULE REQUIREMENT

Clause 6.18.2(b)(6) of the NER requires that a pricing proposal must set out how charges for transmission use of system services are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year. In addition, clause 6.18.7 states that recovery of transmission revenue should not exceed the estimated amount of these charges for the relevant regulatory year, once the overs and unders account has been taken into account.

Ausgrid's Transmission Use of System (TUOS) tariffs are designed to recover the allowed revenue for our transmission network, pass through the TransGrid transmission cost to customers and to recover (return) an under (over) recovery of transmission revenues in the previous period. The process used by Ausgrid to achieve these TUOS pricing outcomes is summarised below:

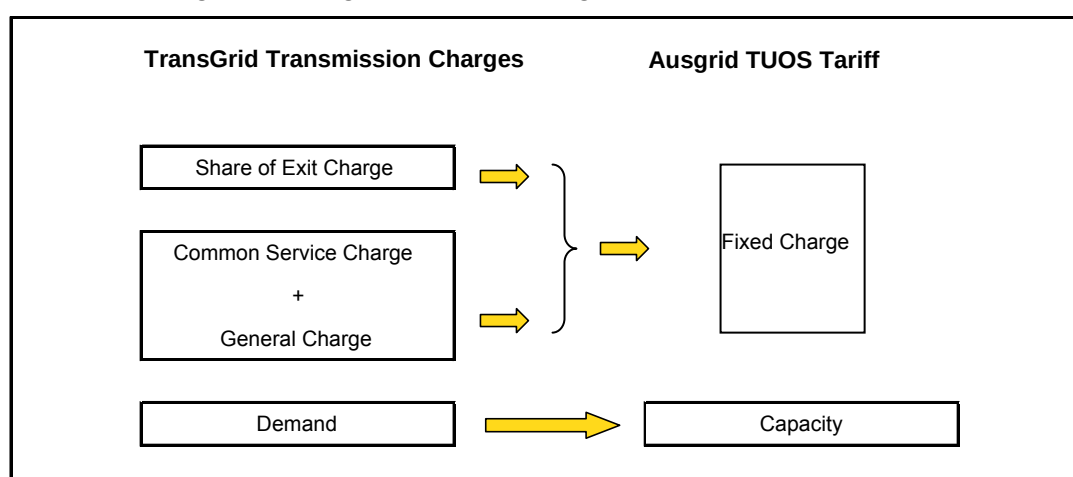
- Step 1: To calculate the annual amount of allowed revenue to recover for Ausgrid transmission assets;
- Step 2: Ausgrid to provide this annual revenue requirement to TransGrid for inclusion in their modelling of transmission charges for the coming year;

- Step 3: TransGrid to provide Ausgrid with the final transmission charges for the coming year in early April, as per the Memorandum of Understanding between the two organisations.
- Step 4: Ausgrid to set TUOS tariffs for the coming year to recover the forecast transmission cost plus the opening balance (positive or negative) of the transmission overs and unders account.

Ausgrid's approach to setting Transmission Use Of System (TUOS) prices is based on the fundamental principle of preserving the TransGrid transmission price signal, where it is desirable to do so from an economic and equity perspective. Given the location-basis of TransGrid's transmission charges, it is only possible to preserve the transmission price signal in the TUOS component of network tariffs for Cost Reflective Network Price (CRNP) customers given the site-specific nature of these network tariffs.

As illustrated in the following figure, Ausgrid's approach to setting TUOS tariffs for CRNP customers preserves the TransGrid transmission price signal, in spite of the structural differences between the two price signals.

Figure 5: Ausgrid's TUOS Pricing Structure for CRNP Customers



While it is not possible to provide individual customers on a published network tariff with cost reflective TUOS prices due to “postage

stamp” basis of these tariffs, it is possible, however, to allocate annual TUOS cost (net of CRNP TUOS recovery) to each tariff class on

an equitable basis using cost allocation principles. As with any cost allocation process, this approach requires a significant degree of averaging of costs and assumptions to be made in relation to the time of use consumption and demand characteristics of tariff classes where interval consumption data does not exist.

While Ausgrid endeavours to set TUOS tariffs consistent with the cost reflectivity principle, this may not always be possible in practice given the following considerations:

- It is not possible to preserve the TransGrid¹⁴ transmission price signal for customers on published network tariffs;
- The likelihood of forecast volume error;
- The NER requirement to set TUOS tariffs to achieve a zero forecast balance of the transmission overs and unders account by the end of the year that the new prices apply; and
- Ausgrid's commitment to transitioning TUOS prices to cost reflective levels, where necessary to avoid imposing unacceptable price shocks on individual customers, as discussed below.

While it is important from an economic perspective to preserve the TransGrid price signal to these customers, Ausgrid believes that it is unfair to burden any individual customer with unacceptable price shocks arising from unexpected changes in the structure of prices. It is for this reason that Ausgrid has decided to transition TUOS tariffs to cost reflective levels over a reasonable period for a number of CRNP customers.

Unders and overs account

The over and unders transmission account is a mechanism to ensure that if transmission revenue recovery is too high or too low in a given regulatory year, Distribution Network Service Providers, such as Ausgrid can recover the difference between actual transmission revenue and transmission payments by adjusting TUOS prices in the subsequent year. Fluctuations in revenue recovery are most likely to be caused by the impact of weather on electricity consumption and demand forecasts.

As required under the NER, Ausgrid have set TUOS tariffs to achieve a forecast zero balance of the overs and unders account for transmission by the end of FY14, as shown in the following table:

¹⁴ Note that TransGrid is the co-ordinating TNSP for the setting of Transmission charges in NSW

Table 12: Overs and Unders Account Forecast Closing Balance – Transmission-Related

Financial Year Ending	Units	Period t-2	Period t-1	Period t
		Audited FY12	Expected FY13	Forecast FY14
Interest rate applicable to balance	%	10.02	10.02	10.02
Opening balance	\$'000	10,118	3,949	(18,912)
Interest on opening balance (365 days)	\$'000	1,014	396	(1,895)
Forecast over/under recover for financial year	\$'000	(6,840)	(22,173)	19,837
Interest charged on over/under recovery for financial year	\$'000	(343)	(1,084)	970
Closing balance of transmission overs/unders account	\$'000	3,949	(18,912)	0.0

8. Changes from the Previous Regulatory Year

RULE REQUIREMENT

Clause 6.18.2(b)(8) of the NER requires that a pricing proposal must describe the nature and extent of change from the previous regulatory year and demonstrate that the changes comply with the Rules and any applicable distribution determination

This chapter sets out the principal changes since the previous regulatory year, and demonstrates that these changes comply with the Rules and the AER's distribution and transmission determinations released on 30 April 2009.

Description of the nature and extent of change from previous regulatory year

Ausgrid proposes to make no changes to tariff structure or to introduce a new published tariff or CRNP tariff in FY 2014. However, Ausgrid has included draft pricing trial tariffs in our proposal for FY 2014. These trial tariffs relate to the Smart City Smart Grid initiative and a potential Ausgrid demand management trial currently under investigation

Unlike in previous years, there will be no mandated transfers of residential customers from IBT to TOU tariff. Importantly, Ausgrid only proposes to transfer a small number of customers between LV TOU capacity tariffs in FY 2014, in accordance with the requirements of our pricing policy (ES7) document.

It is also important to note that Ausgrid proposes in FY 2014 to reduce DUOS prices under the WAPC by around 1.5%, rather than increase these prices by 1% as allowed under the current Determination. This proposal complies with the NER because Ausgrid believes that the economic benefits of returning revenue to customers in the form of lower than allowed DUOS prices under the WAPC is likely to outweigh the associated economic costs.

Demonstration of compliance with the Rules and Determination

The proposed new network tariffs comply with the NER, as discussed in more detail in Chapter 3 of part 2 of this document. The proposed new network tariffs relating to the pricing trials have been submitted in draft form and may require adjustment during the course

of FY14, as discussed in Chapter 5 of part 2 of this document.

In general terms, the pricing proposal demonstrates compliance with the X factor¹⁵ and Price limit under the WAPC by providing the AER with a compliance spreadsheet that sets out the detailed assumptions used to apply the weighted average price cap (WAPC). These assumptions include:

- Audited usage volumes and customer numbers for the year ending June 2012
- adjustments to usage volumes and customer numbers for reasonable estimates
- Distribution use of system (DUOS), transmission use of system (TUOS) and Climate Change Fund prices by tariff component for the year ending June 2013
- Forecast revenues by tariff and tariff class for the years ending June 2013 and 2014

As required, Ausgrid engaged an auditor to undertake an independent review of the WAPC variables included in the Schedules. The statement from PriceWaterhouse Coopers is provided in Attachment I.

¹⁵ Proposed weighted average DUOS prices in FY 2014 increase by less than CPI-X-D constraint under the WAPC.

9. Public Lighting Service

Ausgrid's regulated services are known as "direct control services" meaning that they are subject to a distribution determination. Direct control services are further classified as "standard" control services and "alternative control services". Public lighting services are classified as alternative control services and are subject to a different control mechanism to general network services which are classified as standard control services.

The AER made its final decision for Ausgrid's public lighting services on 15 April 2010. That decision applies a control mechanism from which the prices or tariffs to be applied to each of Ausgrid's public lighting customer are derived. Whilst Clause 6.18 of the Transitional Chapter 6 Rules, applies to Ausgrid's public lighting services the application is limited given that the tariffs can be derived directly from the AER's decision. This section explains the application of these pricing rules to Ausgrid's public lighting services in the context of the decision which has been made by the AER.

Tariffs and tariff classes

All public lighting customers are subject to the tariffs in the AER April 2010 determination and therefore are defined to be a single tariff class.¹⁶ The tariffs that apply to these customers are:

- A fixed capital charge for installed public lighting assets. The fixed capital charges have been provided in confidential Attachment B because they are individual customer charges. This tariff only relates to the capital charges for public lighting assets constructed before 1 July 2009.
- An annuity price per asset type for all new assets. These tariffs and charging parameters are contained in Attachment C. This price relates to the capital charge for new public lighting assets constructed after 30 June 2009, but does not apply to assets where the customer has funded the original construction cost.
- A maintenance price per asset. This tariff relates to the maintenance of

each of Ausgrid's public lights. These tariffs and charging parameters are contained in Attachment C.

Adjustments to prices

The AER April 2010 public lighting decision and clause 6.18.2(b)(5) of the transitional Rules require Ausgrid to explain any adjustment to prices that could occur during the regulatory year. The prices to be levied in the 2013/14 year will not change during the financial year.

In preparation of the 2013/14 prices, Ausgrid has adjusted the fixed capital charges as in 2012/13 some customers paid residual values for assets that they elected to have replaced before the end of their economic life. As a result Ausgrid has deducted this amount from the closing RAB value as of 30 June 2011. The total amount of the residual values accrued in 2012/13 is small (\$145,000), therefore the adjustment is a minor reduction in the fixed capital charge for the relevant customers.

All FY14 public lighting prices were adjusted for CPI (for the four quarters to December 2012), as per page 43 of the AER's determination.

Other requirements of the determination

The AER's final determination for public lighting states that Ausgrid should provide actual public lighting charges for the current year in the pricing proposal.¹⁷ FY13 charges are not yet known and can be provided after the year has ended. Attachments B and C show the public lighting charges proposed for FY14.

National Electricity Rule requirements

The transitional National Electricity Rules require Ausgrid to demonstrate that the expected revenue for each tariff class be between the standalone and avoidable cost of providing the service. The Rules also require Ausgrid to take into account the long run marginal cost for each charging parameter, and have regard for transaction costs and whether relevant tariff class is able to respond to price signals. These Rule requirements

¹⁶ Tariff class: A class of customers for one or more *direct control services* who are subject to a particular tariff or particular tariffs.

¹⁷ AER, EnergyAustralia distribution determination 2009-10 to 2013-14, Alternative control (public lighting) services, 13 April 2010, page 54.

should be considered satisfied for public lighting since Ausgrid follows the requirements of the determination, which was based on the AER's view of efficient cost.

Another transitional Rule that is applicable to direct control services is the assignment of customers to tariff classes (clause 6.18.4). As public lighting is its own tariff class and there have been no-assignment of customers, this Rule obligation has been met for public lighting services. Since all public lighting customers are subject to the same three public lighting tariffs, Ausgrid is satisfied that the criterion in transitional Rule 6.18.4 has been met.

Table 13: Maximum public lighting price increases

Public lighting charging component	Maximum component increase %
Assets installed before July 2009	8.95%
Assets installed post June 2009	1.76%
Maintenance	2.07%

10. Proposed New Network Tariffs

Ausgrid does not propose to introduce any new published or CRNP tariffs in FY 2014. However, Ausgrid has included draft pricing trial tariffs in our pricing proposal for FY14. These trial tariffs relate to an Ausgrid demand management trial currently under investigation and the *Smart Grid Smart City* initiative (refer to Part 1 of this document). Importantly, these tariffs do not require a reasonable estimate of the quantities and customer numbers under the WAPC due to the voluntary natures of these tariffs.

11. Reasonable Estimates

DETERMINATION REQUIREMENT

Appendix J of the AER Determination states that when assessing the reasonableness of the quantity estimates provided by a NSW DNSP, the AER will take the following information into account:

(1) the actual audited quantities sold in relevant units under the origin tariff in previous years;

(2) a forecast of the number of distribution customers that the DNSP states will move to the new tariff/tariff components, and the reasons for the move;

(3) a forecast of the number of distribution customers that the DNSP expects will remain on the origin tariff;

(4) a forecast of the quantities that the DNSP expects will be sold, in relevant units, to those distribution customers that are moved to the new tariff/tariff component;

(5) a forecast of the quantities that the DNSP expects will be sold, in relevant units, to those distribution customers that will remain on the origin tariff;

(6) a forecast of the distribution tariff, and associate revenue, the DNSP expects will be payable by those distribution customers that will be moved to the new tariff/tariff components;

(7) a forecast of the distribution tariff, and associate revenue, the DNSP expects will be payable by those distribution customers that will remain on the origin tariff;

(8) the approach the DNSP used to determine its forecast (for 2-7 above)

(9) the materiality of the reasonable estimates; and

(10) further information as required by the AER.

The purpose of this chapter is to provide a high-level understanding of the concept of reasonable estimates under the WAPC and the Ausgrid approach to calculating reasonable estimates in accordance with the requirements of Appendix J of the Determination. It is important to note that Ausgrid's compliance with Appendix J is demonstrated by the provision of a compliance spreadsheet to the AER, rather than the information provided below.

The concept of a Reasonable Estimate

As explained in Appendix J of the Determination, a DNSP may adopt a pricing strategy that involves:

- The introduction of a new tariff or tariff component.
- The adjustment to an existing tariff component, such as changing the peak period definition.
- The transfer of customers between network tariffs.

Given that both the WAPC and side constraint are based on the audited historical quantities, the new tariffs/tariff components and customer transfers will not be reflected in the historical quantities for two years. Therefore, the historical audited quantities will need to be adjusted to take account of these changes.

Methodology for Calculation of reasonable Estimates

As explained in Appendix J of the Determination, Ausgrid is required for compliance purposes to calculate reasonable estimates under the WAPC on the following basis:

- To exclude voluntary customer transfers arising from the actions of the customer (or the retailer).
- To ensure that the quantity adjustments made due to reasonable estimates relating to tariff changes in period t and period t-1 do not result in the total annual quantities being different to the total audited quantities in period t-2.¹⁸
- To use the origin tariff for period t-1, where both the origin and new tariff are measured in the same units of measure. If there is no origin tariff/tariff component with the same units of measure, the tariff for period t-1 will be set to zero.

The reasonable estimates included in the pricing proposal for FY14 comply with the above requirements given that the quantity

¹⁸ Note - the reasonable estimate of the consumption applying to the new tariff or tariff component is not adjusted for the price elasticity effect.

adjustments under reasonable estimate are based on the historical consumption interval data at the customer level in period t-2. If no historical interval data is available, (ie where only basic metering is installed in the customer's premise), the reasonable estimate of the quantity adjustment will be based on a representative sample of similar customers with historical interval consumption available in period t-2.

Customer transfers in the FY14 Pricing Proposal

The assignment or re-assignment of customers is an important part of our network pricing strategy because it ensures that our customers are on the most appropriate network tariff given the level of their energy consumption and their metering functionality.

Ausgrid proposes to make no customer transfers in FY14, but are required under the NER to make reasonable estimates relating to the small number of customers that were on the Low Voltage TOU Capacity (transitional) tariff at the beginning of FY13.

New tariffs in the FY14 Pricing Proposal

Ausgrid does not propose to introduce any new published or CRNP tariffs in FY 2014. However, Ausgrid has included draft pricing trial tariffs in our pricing proposal for FY14. These trial tariffs relate to an Ausgrid demand management trial currently under investigation and the *Smart Grid Smart City* initiative (refer to Part 1 of this document). Importantly, these tariffs do not require a reasonable estimate of the quantities and customer numbers under the WAPC due to the voluntary natures of these tariffs.¹⁹

Tariff Structure Changes in the FY12 Pricing Proposal

As mentioned previously, Ausgrid proposes to make no changes to the structure of the published and CRNP tariffs in FY 2014. Under the NER, Ausgrid is required to make reasonable estimates of the quantity adjustments associated with the introduction of the three block IBT structure on 1 July 2012.

Time of Use Customer Transfers

Ausgrid does not propose to transfer residential customers from IBT to a TOU tariff in FY 2014. Ausgrid intends to continue to install Type 5 metering in new and upgraded sites. As a result, Ausgrid has not included any residential tariff transfers in the reasonable estimates for FY14.

Forecast DUOS revenue associated with customer transfers

Appendix J of the Determination also requires that the DNSP provides details of the forecast DUOS revenue to be earned by customers under the new tariff compared with the revenue earned under the origin tariff.

A summary of the forecast DUOS revenue impact of the proposed mandated customer transfers between network tariffs has been provided in Appendix H of this document.

¹⁹ Appendix J of the Determination explicitly prohibits reasonable estimates to be made in the case of voluntary customer transfers.

12. Attachments

Attachments have been included with this pricing proposal as follows:

Attachment	Disclosure	Description
A	CONFIDENTIAL	Cost Reflective Network Price (CRNP) Customer Prices
B	CONFIDENTIAL	Customer Bill Information for Public Lighting Customers
C		Price Information for Public Lighting Tariffs
D	CONFIDENTIAL	Completed Compliance Spreadsheet
E		Price and Tariff Class Information for Miscellaneous & Monopoly Charges and Emergency Recoverable Works
F	CONFIDENTIAL	Notification of Climate Change Fund Contribution
G		Procedure on Assignment and Re-assignment of customers to tariffs
H	CONFIDENTIAL	Reasonable Estimates Under the WAPC
I	CONFIDENTIAL	Statement from PriceWaterhouseCoopers Australia
J		TransGrid's transmission prices for FY13
K	CONFIDENTIAL	Economic Cost Model